

Design an 8 kW thermal air-source heat pump with COP 3

Class heat_pump	Feasibility feasible	BoM Readiness Ready	Score 7.7
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Executive Summary

Metric	Value	Unit	Confidence	Notes
Annual useful heat output	16000	kWh/year	medium	Thermal output multiplied by assumed annual full-load hours until project load profile is known.
Estimated CAPEX	0	GBP	low	No CAPEX claim admitted yet; supplier-backed BoM evidence is required before costs appear.
Estimated annual OPEX	0	GBP/year	low	No OPEX claim admitted yet because CAPEX has no sourced BoM basis.

Brief Intake Gate

brief_ready_for_architecture

6/6 intake areas pass; 0 review, 0 blocked. Class heat_pump (high); requirements 2; numeric facts 2; requirement trace 2/2 covered; scratch grammar yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_text	pass	1	211 non-whitespace character(s) in the source brief.	none	Brief text is present.
requirement_quantification	pass	1	2 quantified requirement(s), 2 numeric fact(s).	none	Quantified requirements are sufficient for a first architecture pass.
product_class_selection	pass	1	Selected heat_pump with high confidence.	none	Product class selection is strong enough for scratch architecture.
scratch_design_support	pass	1	heat_pump has a deep scratch architecture grammar.	none	Use scratch universal architecture flow for this brief.
requirement_trace_seed	pass	1	2 covered, 0 partial, 0 uncovered requirement trace row(s).	none	Extracted requirements have trace seeds into the architecture.
assumption_boundary	pass	1	1 brief assumption note(s); no-quantified-default path: no.	none	Assumption boundary is explicit enough for first-pass architecture.

Brief Clarification Plan

no_clarification_needed

No clarification questions are needed before scratch architecture generation.

Architecture Admission Gate

architecture_generation_admitted

6/6 admission areas pass; 0 review, 0 blocked. Can use for review: yes; can proceed to BoM: yes. Source scratch_universal_architecture; next action: none.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_intake_clearance	pass	1	6/6 intake areas pass; class heat_pump (high); 2 quantified requirement(s).	none	Brief intake is strong enough to admit scratch architecture generation.
clarification_clearance	pass	1	0 clarification question(s); 0 required, 0 recommended, 0 architecture-blocking.	none	No brief clarification is needed before architecture generation.
class_grammar_clearance	pass	1	Product class heat_pump; scratch universal grammar supported: yes.	none	Route this class through the scratch universal module grammar.
scratch_stage_clearance	pass	1	Architecture stage passed; source scratch_universal_architecture; modules 10; submodules 31; components 124.	none	Architecture was produced by the scratch universal architecture stage.
architecture_readiness_boundary	pass	1	Architecture ready for BoM: yes; blocking issues 0; required links 4/4.	none	Deterministic architecture readiness checks passed; sourcing work may begin only with explicit evidence.
bom_provenance_hold	pass	1	0 admitted sourcing evidence row(s), 0 source-backed BoM claim(s), 0 provenance violation(s).	none	BoM fields remain behind the explicit sourcing-evidence boundary.

Stage Integrity Gate

stage_trace_accepted

5/5 stage-integrity areas pass; 0 review, 0 blocked. Stages 8/8; ordered: yes; architecture source scratch_universal_architecture; admitted priced lines 0; provenance violations 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
stage_sequence	pass	1	8/8 expected compiler stages are present in order.	none	Stage sequence matches the scratch compiler contract.
stage_payloads	pass	1	8/8 stages have metrics; 8/8 have evidence or explicit limitation; 8/8 have limitations.	none	Every stage carries metrics, evidence or explicit limitation, and limitations.
scratch_architecture_source	pass	1	Product class heat_pump; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Architecture came from the scratch universal grammar.
architecture_before_sourcing	pass	1	Architecture ready: yes; readiness stage passed; sourcing stage warning.	none	Sourcing stage is downstream of architecture readiness and has a consistent status.
sourcing_provenance_boundary	pass	1	0 admitted source evidence row(s), 0 admitted priced line(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN and cost fields stay behind admitted sourcing evidence.

Scratch Lineage Gate

scratch_lineage_clean

6/6 lineage areas pass; 0 review, 0 blocked. Architecture source scratch_universal_architecture; design refs 270; source refs 0; forbidden refs 0; forbidden stage mentions 0; chain-v2 benchmark no (prototype internal depth target).

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_source	pass	1	Product class heat_pump; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Design content came from the scratch universal architecture stage.
design_provenance_refs	pass	1	525 provenance ref(s); 270 design/architecture ref(s); 0 invalid kind(s); 0 forbidden ref(s).	none	All design provenance references are in the allowed scratch provenance vocabulary.
bom_candidate_provenance	pass	1	124 candidate BoM line(s); 0 admitted source evidence row(s); 0 unprovenanced supplier/manufacturer/MPN/cost claim(s).	none	Candidate BoM fields remain unpriced or explicitly source-backed.
source_evidence_boundary	pass	1	0 source provenance ref(s); 0 outside admitted evidence boundary; 0 inside architecture content.	none	Source refs appear only in admitted source evidence, source ledger entries or matching source-backed BoM lines.
chain_v2_quarantine	pass	1	0 forbidden provenance ref(s); 0 forbidden stage-trace mention(s).	none	No chain-v2/state.json/modifier lineage appears inside generated design content or stage traces.
benchmark_isolation	pass	1	Benchmark source: prototype internal depth target; chain benchmark used: no; policy: Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.	none	Benchmark does not use chain-v2 input.

Architecture Freeze Gate

architecture_freeze_review_required

2/6 freeze areas pass; 4 review, 0 blocked. Modules/submodules/components 10/31/124. Structurally ready for sourcing yes; independent review accepted no. Module engineering module_engineering_review_required; interfaces interface_review_ready; requirements coverage_review_ready; next action: Treat the architecture as review-ready until module review questions and critical sourcing blocks close..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
admission_and_lineage	pass	1	Architecture admission architecture_generation_admitted; stage integrity stage_trace_accepted; scratch lineage scratch_lineage_clean.	none	Architecture was admitted from the scratch flow with clean stage and lineage evidence.
module_structure	review	0.76	Component allocation allocation_ready; module engineering module_engineering_review_required; 0/10 modules pass; 5 modules still have critical sourcing blocks.	none	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.
interface_contracts	review	1	Interface verification interface_review_ready; 4/4 required interfaces have carriers; 0 accepted.	none	Collect or accept interface-review evidence before calling the architecture fully frozen.
requirement_coverage	review	1	Requirement coverage coverage_review_ready; 2/2 requirements structurally ready or accepted; accepted ratio 0.	none	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.
engineering_review_state	review	0.83	Pre-BoM engineering engineering_review_ready; 5/6 areas pass; calculations needing review 0; assumptions needing review 18.	none	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.
sourcing_boundary	pass	1	124 candidate BoM lines; 0 admitted sourcing evidence rows; 0 provenance violation(s); 36 critical missing source claim(s).	none	Sourcing and BoM claims remain behind explicit admitted source evidence.

Architecture Freeze Closure Plan

architecture_freeze_review_required

4 closure row(s): 4 ready, 0 blocked. Queues S/E/V/R 1/1/2/0; next row freeze-close:1:module_structure. Structurally ready for sourcing yes; independent review accepted no.

#	Area	Queue	Priority	Status	Required Evidence	Acceptance Criteria	Input Artifacts
1	module_structure	sourcing_intake	major	ready	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.	Module engineering verdict is module_engineering_ready. No module has blocked allocation or unresolved critical sourcing blockers. Every module review question is accepted, revised or explicitly deferred.	*.module-engineering-gate.csv; *.component-allocation-gate.csv; *.engineering-review-pack.csv; *.sourcing-intake-template.csv
2	interface_contracts	verification_intake	minor	ready	Collect or accept interface-review evidence before calling the architecture fully frozen.	Interface verification verdict is accepted_interfaces. Every required interface has endpoint carriers and accepted reviewer evidence.	*.interface-contracts.json; *.interface-verification-gate.csv; *.verification-intake-template.csv
3	requirement_coverage	verification_intake	minor	ready	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.	Requirement coverage verdict is accepted_evidence. Every parsed requirement has architecture links and accepted verification evidence.	*.requirement-coverage-gate.csv; *.engineering-assurance-matrix.csv; *.verification-intake-template.csv
4	engineering_review_state	engineering_review	major	ready	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.	Pre-BoM engineering verdict is engineering_accepted. Calculations, assumptions and verification activities are accepted or explicitly deferred.	*.pre-bom-engineering-gate.csv; *.engineering-calculations.csv; *.engineering-assumptions.csv; *.verification-plan.csv

component_candidates_ready_for_sourcing

7/7 candidate areas pass; 0 review, 0 blocked. Candidate lines 124; critical 9/9; worklist rows 124; duplicate groups 0; provenance violations 0; ready for sourcing yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
candidate_presence	pass	1	124 candidate BoM line(s), 124 distinct component identity key(s).	none	Component candidates exist for downstream sourcing review.
candidate_identity	pass	1	124/124 candidate line(s) have non-generic identity, description and role context.	none	Candidate identities are specific enough for first-pass sourcing worklists.
quantity_basis	pass	1	124/124 candidate line(s) have positive quantity and unit.	none	Candidate quantities have positive values and explicit units.
critical_candidate_coverage	pass	1	9/9 class-critical part(s) are represented by candidate lines; 9 critical BoM line(s).	none	Class-critical parts have candidate lines before sourcing begins.
duplicate_identity_review	pass	1	0 duplicate component identity group(s), 0 involving critical lines.	none	No duplicate component identities need sourcing review.
sourcing_worklist_alignment	pass	1	124/124 unpriced candidate line(s) appear in the sourcing worklist.	none	Every unpriced candidate line is visible in the sourcing worklist.
provenance_boundary	pass	1	0 source evidence row(s), 0 source-backed claim(s), 0 provenance violation(s).	none	Candidate sourcing fields remain clean unless admitted evidence supplies them.

Sourcing Authorization Gate

sourcing_authorized

6/6 sourcing-authorization areas pass; 0 review, 0 blocked. Critical intake 9/9; full intake 124/124; admitted evidence 0; rejected evidence 0; provenance violations 0; authorized yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_authorization	pass	1	architecture_generation_admitted; review yes, BoM yes.	none	Architecture admission permits evidence-gated sourcing work.
component_candidate_authorization	pass	1	component_candidates_ready_for_sourcing; 124 candidate line(s), 124 sourcing-worklist row(s), 0 provenance violation(s).	none	Component candidates are concrete enough for sourcing intake.
critical_intake_scope	pass	1	9/9 critical unpriced line(s) appear in the critical sourcing intake template.	none	Critical sourcing intake scope is complete.
full_worklist_scope	pass	1	124/124 total unpriced line(s) appear in the full sourcing intake template.	none	Full sourcing intake scope matches the unpriced worklist.
evidence_admission_boundary	pass	1	0 admitted source evidence row(s), 0 rejected source evidence row(s).	none	No rejected sourcing evidence is waiting for repair.
provenance_boundary	pass	1	0 source-backed BoM claim(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN, lead-time and cost fields remain behind admitted evidence.

BoM Admission Gate

candidate_bom_authorized

Display mode candidate_only. 2/7 BoM-admission areas pass; 5 review, 0 blocked. Critical priced 0/9; source-backed claims 0; provenance violations 0. Candidate BoM yes; priced BoM no; procurement use no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_authorization	pass	1	sourcing_authorized; authorized yes; 6/6 areas pass.	none	Sourcing is authorized, so the BoM may be shown as a candidate sourcing queue.
candidate_bom_scope	pass	1	124 candidate BoM line(s), 9 critical line(s).	none	Candidate BoM scope exists for sourcing and evidence collection.
source_admission_state	review	0	0 admitted source row(s), 0 rejected source row(s), 0/124 priced line(s).	none	No source rows are admitted; keep this as a candidate-only BoM.
critical_bom_admission	review	0	0/9 critical line(s) have admitted unit costs; 9 critical line(s) remain unpriced.	none	Start with critical sourcing intake before treating any BoM cost as admitted.
source_reference_quality	review	0	no_sourcing_evidence; 0/0 source-quality rows pass; 0 protocol fixture row(s).	none	No source evidence is present; keep the BoM candidate-only.
provenance_boundary	review	0.94	0 source-backed claim(s), 36 critical missing-source claim(s), 0 provenance violation(s).	none	Missing critical source claims are acceptable only in candidate-only mode; collect evidence before priced BoM use.
display_mode	review	0.25	Display mode will be candidate_only.	none	Show candidate BoM as a sourcing queue with supplier, MPN and cost fields empty.

Report Readiness Gate

architecture_review_ready

5/6 sections at or above 8. Mean score 7.67. Architecture ready: yes. Critical unpriced lines: 9. Verification accepted: 0/25; unaccepted: 25. Verification acceptance ratio: 0.

- bom score 1 is below 8.
- 9 critical BoM line(s) lack source-backed cost evidence.
- 5 verification activity/activities remain blocked.
- 0/25 non-sourcing verification activity/activities have accepted reviewer evidence.

Section	Score	Target	B/M/m Issues	Action
executive_summary	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
brief_requirements	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
design_modules	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
bom	1	below target	9/1/0	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates.
feasibility_notes	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
sources_references	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.

Pre-BoM Engineering Gate

engineering_review_ready

5/6 areas pass; 1 review, 0 blocked. Modules/submodules/components: 10/31/124. Calculations within envelope 2/2; calculation blockers 0. Assumption review rows 18; architecture blockers 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	10 modules, 31 submodules and 124 component words pass architecture readiness.	none	Architecture is structurally ready for engineering review.
component_allocation	pass	1	31/31 submodules ready; 9/9 critical parts allocated; 0 duplicate component groups.	none	Component candidates are allocated well enough for engineering review.
interface_verification	pass	1	4/4 required interfaces have endpoint carriers; 0 accepted with evidence.	none	Required interfaces are structurally ready for reviewer evidence.
requirement_coverage	pass	1	2/2 requirements are structurally ready or accepted; accepted evidence ratio 0.	none	Requirements are structurally ready for reviewer evidence.
calculation_envelope	pass	1	2/2 calculations are within envelope; 0 need review; 0 block.	none	Calculation envelope has no deterministic blocker.
engineering_assumptions	review	0.25	6/24 assumptions are brief-supported or model-present; 18 need review/source evidence; 0 block architecture.	none	Review assumptions and keep BoM/source-required assumptions out of trusted claims until evidence is admitted.

BoM Costing Gate

costing_not_started

1/7 costing areas pass; 0 review, 6 blocked. Critical priced lines 0/9; unpriced critical lines 9. Source-backed claims 0; provenance violations 0; sourcing evidence 0/0 production-ready, 0 protocol. Source quality: no_sourcing_evidence (0/0 pass, 0 placeholder URL, 0 quote-anchored). BoM total £0; CAPEX £0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_admission	blocked	0	0/124 BoM lines have admitted pricing; rejected source records 0.	energy_conversion_transduction_compressor_inverter_stage_inverter_scroll_compressor: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_conversion_transduction_expansion_metering_stage_electronic_expansion_valve: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. environmental_interface_outdoor_evaporator_outdoor_finned_evaporator_coil: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. environmental_interface_fan_air_path_variable_speed_fan_assembly: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. mass_fluid_transport_process_condenser_heat_exchanger_brazed_plate_heat_exchanger: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 4 more	Admit source-backed supplier, manufacturer, MPN and unit-cost evidence before costing review.
critical_line_pricing	blocked	0	0/9 critical lines have admitted unit costs.	energy_conversion_transduction_compressor_inverter_stage_inverter_scroll_compressor: Inverter scroll compressor energy_conversion_transduction_expansion_metering_stage_electronic_expansion_valve: Electronic expansion valve environmental_interface_outdoor_evapo	Price every critical line from admitted source evidence.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				rator_outdoor_finne d_evaporator_coil: Outdoor finned evaporator coil environmental_inter face_fan_air_path_ variable_speed_fan _assembly: Variable-speed fan assembly mass_fluid_transpo rt_process_conden ser_heat_exchanger _brazed_plate_heat _exchanger: Brazed plate heat exchanger plus 4 more	
source_backed_pro venance	blocked	0.94	0 source-backed claims; 36 critical missing-source claims; 0 provenance violations.	energy_conversion_ transduction_compr essor_inverter_stag e_inverter_scroll_co mpressor/supplier: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. energy_conversion_ transduction_compr essor_inverter_stag e_inverter_scroll_co mpressor/manufact urer: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. energy_conversion_ transduction_compr essor_inverter_stag e_inverter_scroll_co mpressor/mpn: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. energy_conversion_ transduction_compr essor_inverter_stag e_inverter_scroll_co mpressor/unit_cost _gbp: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. energy_conversion_ transduction_expan sion_metering_stag	Keep supplier, manufacturer, MPN, unit cost and lead- time fields empty until source-backed evidence is admitted.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				e_electronic_expansion_valve/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 31 more	
component_identity	pass	1	0 duplicate component identity group(s); 0 include critical lines.	none	Component identities are unique enough for line-level costing.
sourcing_evidence_authenticity	blocked	0	0/0 sourcing evidence rows are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Add sourcing evidence rows before BoM costing can start.
source_reference_quality	blocked	0	0/0 sourcing refs pass quality checks; 0 placeholder URL, 0 quote-anchored, 0 fresh timestamp.	none	Add source references before costing review.
cost_totals	blocked	0	BoM total 0 GBP; CAPEX 0 GBP; annual OPEX 0 GBP.	9 critical line(s) are unpriced, so totals are incomplete.	Do not use cost totals until every critical line has an admitted unit cost.

Evidence Acquisition Plan

39 missing-evidence row(s): 12 sourcing, 27 verification. Ready 36; blocked 0. Blocks BoM 12; blocks publish 39. Next row: acquire:4:assumption:metric:capex_gbp.

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
1	sourcing	blocker	open	readiness:2	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
2	verification	blocker	open	readiness:3	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
3	verification	blocker	open	readiness:4	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
4	sourcing	blocker	ready_for_intake	assumption:metric:capex_gbp	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						sourceGrade; evidence.ref; evidence.quote; retrievedAt		
5	sourcing	blocker	ready_for_intake	assumption:metric:opex_annual_gbp	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
6	sourcing	blocker	ready_for_intake	inverter_scroll_compressor	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
7	sourcing	blocker	ready_for_intake	electronic_expansion_valve	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.ref ; evidence.quote; retrievedAt		
8	sourcing	blocker	ready_for_in take	outdoor_finned_evaporator_coil	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
9	sourcing	blocker	ready_for_in take	variable_speed_fan_assembly	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
10	sourcing	blocker	ready_for_in take	brazed_plate_heat_exchanger	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.quote; retrievedAt		
11	sourcing	blocker	ready_for_intake	hydronic_circulation_pump	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
12	sourcing	blocker	ready_for_intake	hydronic_safety_valve_kit	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
13	sourcing	blocker	ready_for_intake	heat_pump_controller_pcb	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						ote; retrievedAt		
14	sourcing	blocker	ready_for_intake	refrigerant_sensor_and_pressure_transducers	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture://source	yes
15	verification	major	ready_for_intake	design_review:energy_conversion_transduction	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture://review; source-evidence claims that belong in sourcing intake	no
16	verification	major	ready_for_intake	calculation:energy_conversion_transduction	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture://review; source-evidence claims that belong in sourcing intake	no
17	verification	major	ready_for_intake	design_review:environmental_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture://review; source-evidence claims that	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
							belong in sourcing intake	
18	verification	major	ready_for_intake	calculation: environmental_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
19	verification	major	ready_for_intake	design_review:mass_fluid_transport_process	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
20	verification	major	ready_for_intake	calculation: mass_fluid_transport_process	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
21	verification	major	ready_for_intake	design_review:control_compute_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
22	verification	major	ready_for_intake	calculation: control_compute_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
23	verification	major	ready_for_intake	design_review:sensing_instrumentation	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
24	verification	major	ready_for_intake	calculation: sensing_instrumentation	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
25	verification	major	ready_for_intake	design_review:power_distribution	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
26	verification	major	ready_for_intake	design_review:structure_containment	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
27	verification	major	ready_for_intake	design_review:safety_protection	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
28	verification	major	ready_for_intake	design_review:maintenance_serviceability	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
29	verification	major	ready_for_intake	design_review:hmi_ergonomics	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
30	verification	major	ready_for_intake	interface_review:environmental_interface:ener	named_engineering_review	activityId; evidenceKind; reviewerNa	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
				gy_conversion_transduction:refrigerant_loop		me; verdict; evidenceRef; evidenceNote; reviewedAt	assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	
31	verification	major	ready_for_intake	interface_review:energy_conversion_transduction:mass_flow_id_transport_process:hydraulic_loop	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
32	verification	major	ready_for_intake	interface_review:control_compute_communication:energy_conversion_transduction:control_buses	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
33	verification	major	ready_for_intake	interface_review:sensing_instrumentation:control_computation_communication:sensor_bus	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
34	verification	major	ready_for_intake	compliance_review:EN-14511	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNo	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						te; reviewedAt	evidence claims that belong in sourcing intake	
35	verification	major	ready_for_in take	compliance _review:EN- 378	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
36	verification	major	ready_for_in take	compliance _review:EN- 60335-2- 40	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
37	verification	major	ready_for_in take	risk_review: 1	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
38	verification	major	ready_for_intake	risk_review: 2	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
39	verification	major	ready_for_intake	risk_review: 3	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

Document Trust Gate

evidence_blocked

1/7 trust areas pass; 3 review, 3 blocked. Architecture ready: yes. Claim gate: evidence_blocked. Reviewer evidence 0/25. Source-backed BoM claims 0; critical missing BoM claims 36; provenance violations 0. Evidence authenticity: no_evidence (0/0 production-ready, 0 protocol).

- bom: score 1 below 8
- architecture design has 0 blocked/source-required claim(s) and 41 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom sourcing has 45 blocked/source-required claim(s) and 115 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 6 review-required claim(s).
- energy_conversion_transduction_compressor_inverter_stage_inverter_scroll_compressor/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_conversion_transduction_compressor_inverter_stage_inverter_scroll_compressor/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_conversion_transduction_compressor_inverter_stage_inverter_scroll_compressor/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_conversion_transduction_compressor_inverter_stage_inverter_scroll_compressor/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_conversion_transduction_expansion_metering_stage_electronic_expansion_valve/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_conversion_transduction_expansion_metering_stage_electronic_expansion_valve/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_conversion_transduction_expansion_metering_stage_electronic_expansion_valve/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	10 modules, 31 submodules and 4 required links pass deterministic architecture gates.	none	No deterministic architecture blocker remains; continue with reviewer evidence before treating it as accepted.
section_scores	blocked	0.83	5/6 scored sections are at or above 8; mean 7.67.	bom: score 1 below 8	Repair below-target sections through the specific score-ledger actions.
claim_evidence	blocked	0.01	2/214 claims pass; 47 blocked/source-required, 165 review-required.	architecture design has 0 blocked/source-required claim(s) and 41 review-required claim(s). engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s). bom sourcing has 45 blocked/source-required claim(s) and 115 review-required claim(s). compliance risk has 0 blocked/source-required claim(s) and 6 review-required claim(s).	Accept, correct or reject generated module and submodule allocation claims through engineering review. Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence. Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims. Accept, correct or reject class-pack regulatory and risk claims through compliance/domain review.
requirement_assurance	review	0	0/2 requirements have accepted assurance; 2 need review, 0 blocked, 0 unlinked.	none	Collect reviewer evidence for requirement assurance rows before publication.
reviewer_evidence	review	0	0/25 non-sourcing verification activities accepted; 0 rejected, 0 deferred, 25 pending/blocked.	none	Collect accepted reviewer evidence for every non-sourcing verification activity.
bom_provenance	blocked	0.94	0 source-backed BoM claims, 36 critical missing-source claims, 0 provenance violations.	energy_conversion_transduction_compressor_inverter_stage_inverter_scroll_compressor/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_conversion_transduction_compr	Admit source-backed evidence for critical BoM supplier, manufacturer, MPN and cost claims.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				essor_inverter_stag e_inverter_scroll_co mpressor/manufact urer: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. energy_conversion_ transduction_compr essor_inverter_stag e_inverter_scroll_co mpressor/mpn: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. energy_conversion_ transduction_compr essor_inverter_stag e_inverter_scroll_co mpressor/unit_cost _gbp: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. energy_conversion_ transduction_expan sion_metering_stag e_electronic_expan sion_valve/supplier: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. plus 31 more	
evidence_authentici ty	review	0	0/0 evidence row(s) are production- ready; 0 protocol fixture, 0 review/missing metadata.	none	Admit source and reviewer evidence before trusting the document.

Evidence Authenticity Gate

no_evidence

0/0 evidence rows are production-ready. Protocol fixtures 0; review-required 0; missing metadata 0. External URLs 0; internal references 0; local files 0; unknown references 0.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
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Evidence Replacement Plan

No evidence replacement rows remain for the current authenticity verdict: no_evidence.

Trust Repair Plan

6 package(s): 6 ready, 0 waiting. Next package: sourcing_intake. Closure rows 47; trust blockers 49. Trust verdict: evidence_blocked.

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
2	Admit Source-Backed BoM Claims	ready	claim_evidence; bom_provenance	12	40	Supplier, manufacturer, MPN, unit cost, lead time, retrieved-at timestamp and source reference for critical BoM lines. Rejected or duplicate component identities resolved before pricing shared items.	*.sourcing-intake-template.csv; *.sourcing-pack.csv; *.bom-provenance-manifest.csv; *.component-identity.csv	BoM provenance has zero critical missing-source claims. Claim Evidence Gate BoM sourcing area is no longer evidence_blocked.
3	Resolve Engineering Review Questions	ready	claim_evidence; requirement_assurance	7	4	Accepted/corrected/rejected answers to review-pack questions. Calculation and assumption decisions tied to reviewer evidence references.	*.engineering-review-pack.csv; *.engineering-assurance-matrix.csv; *.engineering-assumptions.csv; *.engineering-calculations.csv	Requirement assurance rows have accepted or explicitly deferred engineering review evidence. Generated architecture and calculation claims are accepted, corrected or rejected.
4	Admit Reviewer Verification Evidence	ready	reviewer_evidence	27	0	Accepted, rejected or deferred verification evidence for every non-source verification activity.	*.verification-intake-template.csv; *.verification-plan.csv; *.verification-ledger.csv	Every non-source verification activity has accepted reviewer evidence or an explicit defer decision.
5	Validate Evidence Authenticity	ready	evidence_authenticity	0	0	Production supplier/catalogue URLs for BoM source records. Named reviewer evidence references that are not protocol fixtures.	*.evidence-authenticity-gate.csv; *.sourcing-ledger.csv; *.verification-ledger.csv; *.claim-ledger.csv	Evidence Authenticity Gate verdict is production_ready. No protocol_fixture, unknown_reference or missing_metadata evidence

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
								rows remain.
6	Review Compliance And Risk Claims	ready	claim_evidence	0	4	Compliance -domain review of class-pack standards, hazards and mitigation claims.	*.claim-ledger.csv; *.engineering-review-pack.csv	Compliance and risk claims are accepted, corrected or rejected by a domain reviewer.
7	Repair Residual Section Scores	ready	section_scores	1	1	A rerun after evidence intake, plus score-ledger fixes for any section still below target.	*.score-ledger.csv; *.readiness-gate.csv; *.document-trust-gate.csv	All scored sections are at or above target and Document Trust Gate is no longer blocked by scoring.

Claim Ledger

214 claims: 2 brief-supplied, 0 reviewer-accepted, 0 source-backed, 3 calculated needing review, 162 generated needing review, 47 source-required, 0 blocked. Blocking: 47 BoM, 210 publishable.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	Heat pump controller PCB / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Heat pump controller PCB / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Heat pump controller PCB / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Heat pump controller PCB / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Inverter scroll compressor / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Inverter scroll compressor / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	Inverter scroll compressor / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Inverter scroll compressor / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Electronic expansion valve / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Electronic expansion valve / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Electronic expansion valve / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Electronic expansion valve / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Variable-speed fan assembly / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Variable-speed fan assembly / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Variable-speed fan assembly / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Variable-speed fan assembly / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Outdoor finned evaporator coil / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Outdoor finned evaporator coil / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/ma

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Outdoor finned evaporator coil / supplier	supplier is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Outdoor finned evaporator coil / unit_cost_g bp	unit_cost_g bp is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Hydronic circulation pump / manufactur er	manufactur er is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Hydronic circulation pump / mpn	mpn is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Hydronic circulation pump / supplier	supplier is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Hydronic circulation	unit_cost_g bp is	none	yes	yes	Admit source-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
			pump / unit_cost_gbp	required but not source-backed.				backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Brazed plate heat exchanger / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Brazed plate heat exchanger / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Brazed plate heat exchanger / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Brazed plate heat exchanger / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Hydronic safety valve kit / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	Hydronic safety valve kit / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Hydronic safety valve kit / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Hydronic safety valve kit / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Refrigerant sensor and pressure transducers / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Refrigerant sensor and pressure transducers / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Refrigerant sensor and pressure transducers / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	Refrigerant sensor and pressure transducers / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
component_candidate	source_required	generated_model	Heat pump controller PCB	Heat pump controller PCB is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Inverter scroll compressor	Inverter scroll compressor is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Electronic expansion valve	Electronic expansion valve is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Variable-speed fan assembly	Variable-speed fan assembly is an architecture-derived component candidate	model:scratch_universal_architecture.heat_pump	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				with quantity 1 candidate.				time evidence through sourcing intake.
component_candidate	source_required	generated_model	Outdoor finned evaporator coil	Outdoor finned evaporator coil is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Hydronic circulation pump	Hydronic circulation pump is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Brazed plate heat exchanger	Brazed plate heat exchanger is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Hydronic safety valve kit	Hydronic safety valve kit is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Refrigerant sensor and	Refrigerant sensor and	model:scratch_universal_architecture.heat_pump	yes	yes	Admit source-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
			pressure transducers	pressure transducers is an architecture-derived component candidate with quantity 1 candidate.	l_architecture.heat_pump			backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
headline_metric	source_required	assumption	Estimated CAPEX	Estimated CAPEX = 0 GBP.	formula:capex=sum(bom.totalCostGbp); class_pack:heat_pump.class_pack	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
headline_metric	source_required	assumption	Estimated annual OPEX	Estimated annual OPEX = 0 GBP/year.	formula:opex=capex*0.04	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
component_candidate	generated_needs_review	generated_model	controller enclosure	controller enclosure is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	non-volatile fault memory	non-volatile fault memory is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	real-time clock module	real-time clock module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	compressor contactor	compressor contactor is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	fan PWM output board	fan PWM output board is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	pump relay output	pump relay output is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	safety chain input terminal	safety chain input terminal is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	fault LED stack	fault LED stack is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	installer QR code label	installer QR code label is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	Modbus service port	Modbus service port is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	USB commissioning adapter	USB commissioning adapter is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	compressor inverter drive	compressor inverter drive is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	compressor mounting grommet set	compressor mounting grommet set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	crankcase heater band	crankcase heater band is an architecture-derived component candidate	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	condenser insulation sleeve	with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	installation instance.
				condenser insulation sleeve is an architecture-derived component candidate with quantity 1 candidate.				Review whether this is a unique physical item, shared component or separate installation instance.
				hydronic stub-out pair				Review whether this is a unique physical item, shared component or separate installation instance.
				plate heat exchanger mounting bracket				Review whether this is a unique physical item, shared component or separate installation instance.
				refrigerant stub-out pair				Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	EEV stepper harness	EEV stepper harness is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	liquid-line filter drier	liquid-line filter drier is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	sight glass indicator	sight glass indicator is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	brazed joint sleeve set	brazed joint sleeve set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	copper refrigerant tube set	copper refrigerant tube set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.heat_pump	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Showing first 70 of 214 claims; export CSV for the full claim ledger.

Claim Evidence Gate

evidence_blocked

2/214 claims pass by being brief-supplied, source-backed or reviewer-accepted. 47 blocked/source-required claims, 165 review-required claims. Claim pass ratio 0.01. Blocking: 47 BoM, 210 publishable.

- architecture design has 0 blocked/source-required claim(s) and 41 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom sourcing has 45 blocked/source-required claim(s) and 115 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 6 review-required claim(s).

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
brief_requirements	complete	2	2	0	0	0	0	0	1	No claim-evidence action required for this area.
architecture_design	review	41	0	41	0	0	0	41	0	Accept, correct or reject generated module and submodule allocation claims through engineering review.
engineering_math	blocked	5	0	3	2	2	2	3	0	Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence.
bom_sourcing	blocked	160	0	115	45	45	45	160	0	Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims.
compliance_risk	review	6	0	6	0	0	0	6	0	Accept, correct or reject class-pack regulatory and risk claims through compliance

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
										ce/domain review.

Evidence Gap Register

47 open evidence gaps: 15 blockers, 32 major, 0 minor. Closure paths: 12 sourcing intake, 27 verification intake, 7 engineering review, 0 architecture revision. Blocking rows: architecture 0, BoM 12, publishable 47.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
readiness_blocker	blocker	open	Report readiness gate	bom score 1 is below 8.	score_repair	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	9 critical BoM line(s) lack source-backed cost evidence.	sourcing_intake	bom; publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	5 verification activity/activities remain blocked.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	0/25 non-sourcing verification activity/activities have accepted reviewer evidence.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier,

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							manufacturer, MPN and cost evidence for critical BoM lines.
assumption_review	major	needs review	Annual useful heat output	Annual useful heat output uses thermal_output_kw * annual_full_load_hours with medium confidence.	engineering_review	publishable	Review the formula, input values and operating envelope before treating this metric as a performance claim.
assumption_review	blocker	ready for intake	Estimated CAPEX	Estimated CAPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	blocker	ready for intake	Estimated annual OPEX	Estimated annual OPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	major	needs review	COP And Input Power	COP And Input Power currently evaluates as pass against COP 2.5 to 5.5 for plausible air-source heat-pump operating points.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Refrigerant/Hydronic Closure	Refrigerant/Hydronic Closure currently evaluates as pass against refrigerant loop, hydronic loop, control bus and sensor bus present.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Safety And Serviceability Interfaces	Safety And Serviceability Interfaces currently evaluates as pass against pressure	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				relief, protective earth, condensate drain and service access present.			rationale for this envelope.
assumption_review	major	needs review	EN-14511	Heat pump performance test conditions is relevant to this product class in UK/EU.	engineering_review	publishable	Capacity, COP and operating-point test evidence.
assumption_review	major	needs review	EN-378	Refrigerating systems and heat pumps safety is relevant to this product class in UK/EU.	engineering_review	publishable	Refrigerant charge, pressure protection and safety assessment.
assumption_review	major	needs review	EN-60335-2-40	Electrical safety for heat pumps and air conditioners is relevant to this product class in UK/EU.	engineering_review	publishable	Electrical safety, leakage, temperature and abnormal-operation evidence.
sourcing_evidence	blocker	ready for intake	Inverter scroll compressor	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Electronic expansion valve	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
sourcing_evidence	blocker	ready for intake	Outdoor finned evaporator coil	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Variable-speed fan assembly	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Brazed plate heat exchanger	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Hydronic circulation pump	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.

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Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
sourcing_evidence	blocker	ready for intake	Hydronic safety valve kit	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Heat pump controller PCB	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Refrigerant sensor and pressure transducers	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
verification_evidence	major	ready for intake	Refrigerant Energy Conversion	Review Refrigerant Energy Conversion architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Refrigerant Energy Conversion	Verify Refrigerant Energy Conversion requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Outdoor Air Interface	Review Outdoor Air Interface architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
verification_evidence	major	ready for intake	Outdoor Air Interface	Verify Outdoor Air Interface requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Hydronic Heat Delivery	Review Hydronic Heat Delivery architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Hydronic Heat Delivery	Verify Hydronic Heat Delivery requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Control Compute Communication	Review Control Compute Communication architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Control Compute Communication	Verify Control Compute Communication requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Sensing Instrumentation	Review Sensing Instrumentation architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Sensing Instrumentation	Verify Sensing Instrumentation requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Electrical Power Distribution	Review Electrical Power Distribution architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
verification_evidence	major	ready for intake	Monobloc Structure And Containment	Review Monobloc Structure And Containment architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Safety Protection	Review Safety Protection architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Maintenance And Commissioning	Review Maintenance And Commissioning architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	User And Installer Interface	Review User And Installer Interface architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Outdoor Air Interface -> Refrigerant Energy Conversion	Verify refrigerant_loop interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Refrigerant Energy Conversion -> Hydronic Heat Delivery	Verify hydronic_loop interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Control Compute Communication -> Refrigerant Energy Conversion	Verify control_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Sensing Instrumentation -> Control Compute Communication	Verify sensor_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
verification_evidence	major	ready for intake	System Compliance	Plan evidence for EN-14511	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for EN-378	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for EN-60335-2-40	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Refrigerant leak or flammability hazard	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Frozen outdoor coil reduces capacity	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Hydronic overpressure or low flow	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Closure Plan

4 active phases. Next phase: sourcing_intake. Rows by phase: architecture 0, sourcing 12, engineering 7, verification 27, score repair 1.

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
2	Admit Source-Backed BoM Evidence	12	12/0/0	bom; publishable	Critical BoM costs, suppliers, manufacturers and MPNs must be source-backed before cost claims are credible.	Every critical BoM line has admitted supplier, manufacturer, MPN, unit cost and source reference. BoM provenance manifest has zero critical missing-source claims.	assumption:metric:capex_gbp; assumption:metric:opex_annual_gbp; readiness:2; sourcing:brazed_plate_heat_exchanger; sourcing:electronic_expansion_valve; sourcing:heat_pump_controller_pcb; sourcing:hydraulic_circulation_pump; sourcing:hydraulic_safety_valve_kit; plus 4 more
3	Complete Engineering Review	7	0/7/0	publishable	Deterministic arithmetic can expose claims, but engineering judgment must review envelopes, assumptions and operating cases.	All calculation and assumption review rows are accepted, revised or explicitly deferred with reviewer rationale.	assumption:compliance:EN-14511; assumption:compliance:EN-378; assumption:compliance:EN-60335-2-40; assumption:metric:headline_output; assumption:sanity:heatpump_cop_input_power; assumption:sanity:heatpump_refrigerant_hydraulic_closure; assumption:sanity:heatpump_safety_serviceability
4	Accept Reviewer Evidence	27	2/25/0	publishable	Reviewer acceptance is tracked separately from deterministic scoring and must close before publication.	Every non-source verification activity is accepted, or explicitly deferred and removed from publication criteria.	readiness:3; readiness:4; verification:calculation:control_compute_communication; verification:calculation:energy_conversion_transduction; verification:calculation:environmental_inte

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
							rface; verification:cal culation:mass _fluid_transp ort_process; verification:cal culation:sensi ng_instrumen tation; verification:co mpliance_revi ew:EN-14511; plus 12 more
5	Repair Remaining Section Scores	1	1/0/0	publishable	After evidence intake, rerun scoring to remove residual deterministic section blockers.	Every scored report section is at or above target and readiness verdict is publishable.	readiness:1

Depth Benchmark

prototype internal depth target. Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.

5 meeting benchmark, 2 below benchmark, average comparable ratio 1.03.

Dimension	Scratch	Benchmark	Ratio	Status	Notes	Action
Functional modules	10 modules	9 modules	1.11	meets	Core architecture breadth.	Add missing functional modules only when the product class genuinely requires them.
Submodules	31 submodules	27 submodules	1.15	meets	Submodule count is a rough proxy for engineering decomposition depth.	Expand thin modules with concrete sub-functions and carrier interfaces.
Component candidates	124 components	108 components	1.15	meets	Component-word count is a rough proxy for BoM review surface, not sourcing quality.	Add justified component candidates where submodules still hide assemblies.
Interface link volume	49 links	34 links	1.44	meets	Compares link volume only; chain-v2 grammar links and scratch interface edges are not semantically identical.	Add interface contracts only where they clarify a real energy, material, signal, load or service path.
Admitted priced-line ratio	0 ratio	0.75 ratio	0	below	Prototype target expects most critical lines to become source-backed before publication.	Use sourcing intake to admit source-backed price/manufacturer/MPN evidence; do not copy benchmark estimates.
Sections scoring >=8	5 sections	6 sections	0.83	below	Readiness gate requires every scored section to reach the target.	Clear section-specific readiness actions, especially BoM evidence.
Verification activities	30 activities	20 activities	1.5	meets	Scratch-only evidence workflow depth; not compared to chain-v2 content.	Keep adding verification activities when new requirements, interfaces or standards are introduced.

Brief Requirements

Design an 8 kW thermal air-source heat pump with COP 3.5, inverter compressor, R290 refrigerant, outdoor evaporator coil, plate heat exchanger, hydronic circulation pump, monobloc enclosure, and defrost control.

Requirement	Value	Unit
Thermal output	8	kW
Coefficient of performance	3.5	COP

Requirement Traceability

Requirement	Status	Value	Architecture Coverage	Evaluators	Notes
Thermal output	covered	8 kW	Refrigerant Energy Conversion / Compressor inverter stage; Outdoor Air Interface / Outdoor evaporator; Hydronic Heat Delivery / Condenser heat exchanger	metric:headline_output; sanity:heatpump_cop_input_power; sanity:heatpump_refrigerant_hydronic_closure	Thermal output is traced through refrigerant conversion, outdoor heat pickup and water-side delivery. Requirement has architecture coverage plus metric or engineering sanity coverage.
Coefficient of performance	covered	3.5 COP	Refrigerant Energy Conversion / Compressor inverter stage; Control Compute Communication / Main controller; Sensing Instrumentation / Refrigerant sensing	metric:headline_output; sanity:heatpump_cop_input_power; sanity:heatpump_refrigerant_hydronic_closure	COP is traced through compressor/control behaviour and the sensing needed to validate operating points. Requirement has architecture coverage plus metric or engineering sanity coverage.

Requirement Coverage Gate

coverage_review_ready

2 requirement row(s): 0 accepted with evidence, 2 coverage-ready, 0 needing review, 0 blocked, 0 unlinked. Structural coverage ratio 1; accepted evidence ratio 0.

Requirement	Status	Present Signals	Missing Signals	Modules	Verification Activities	Required Action
Coefficient of performance	review ready	architecture_link ; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	energy_conversion_transduction; control_computation_communication ; sensing_instrumentation	design_review:energy_conversion_transduction; calculation:energy_conversion_transduction; design_review:control_computation_communication; calculation:control_computation_communication; design_review:sensing_instrumentation; calculation:sensing_instrumentation	Collect accepted reviewer evidence; structural requirement coverage is ready for review.

Thermal output	review ready	architecture_link ; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	energy_conversion_transduction; environmental_interface; mass_fluid_transport_process	design_review:energy_conversion_transduction; design_review:environmental_interface; calculation:environmental_interface; design_review:mass_fluid_transport_process; calculation:mass_fluid_transport_process	Collect accepted reviewer evidence; structural requirement coverage is ready for review.
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Engineering Assurance Matrix

2 requirements: 0 accepted, 0 ready, 2 needing review, 0 blocked, 0 unlinked. 2 have calculation links, 2 have review questions, 0 have accepted verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
Coefficient of performance	needs review	covered	Refrigerant Energy Conversion; Control Compute Communication; Sensing Instrumentation	heat_pump_electrical_input_kw; heat_pump_cop_envelope	assumption_resolution: component: control_compute_communication_main_controller_heat_pump_controller_pcb; assumption_resolution: component: energy_conversion_transduction_compressor_inverter_stage_inverter_scroll_compressor; assumption_resolution: component: energy_conversion_transduction_expansion_metering_stage_electronic_expansion_valve; assumption_resolution: component: sensing_instrumentation_refrigerant_sensing_refrigerant_sensor_and_pressure_transducers; assumption_resolution: sanity:heatpump_cop_input_power; plus 20 more	pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.
Thermal output	needs review	covered	Refrigerant Energy Conversion; Outdoor Air Interface; Hydronic Heat Delivery	heat_pump_electrical_input_kw	assumption_resolution: component: energy_conversion_transduction_compressor_inverter_stage_inverter_scroll_compressor; assumption_resolution:	pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
					component: energy_conversion_expansion_metering_stage_electronic_expansion_valve; assumption_resolution: component: environmental_interface_fan_air_path_variable_speed_fan_assembly; assumption_resolution: component: environmental_interface_outdoor_evaporator_outdoor_finned_evaporator_coil; assumption_resolution: component: mass_fluid_transport_process_circulation_pump_group_hydraulic_circulation_pump ; plus 22 more			

Compiler Stage Trace

Stage	Status	Summary	Metrics	Evidence	Limitations
Brief Parsing	passed	Extracted quantified requirements from the user brief before design generation.	requirement_count: 2; numeric_fact_count : 2; requirement_trace_covered: 2; requirement_trace_partial: 0; requirement_trace_uncovered: 0	thermal_output_kw: 8 kW; cop: 3.5 COP; thermal_output_kw: covered; cop: covered	Parsing is deterministic and currently recognises only a small set of hardware requirement patterns.
Product-Class Selection	passed	Selected heat_pump with high confidence from brief keywords.	selected_class: heat_pump; confidence: high; keyword_scores: heat_pump:7	heat_pump: 7	Classifier is keyword based; ambiguous multi-domain briefs need an LLM or richer taxonomy pass.
Universal Module Architecture	passed	Built a 10-module heat_pump architecture from the scratch universal architecture grammar.	module_count: 10; architecture_source : scratch_universal_architecture	Refrigerant Energy Conversion; Outdoor Air Interface; Hydronic Heat Delivery; Control Compute Communication; Sensing Instrumentation; Electrical Power Distribution; plus 4 more	The scratch grammar is hand-authored and deterministic; it still needs independent engineering review.
Submodule Expansion	passed	Expanded each module into engineering submodules with local purpose, interfaces and component candidates.	submodule_count: 31; average_submodules_per_module: 3.1	energy_conversion_transduction.compressor_inverter_stage; energy_conversion_transduction.expansion_metering_stage; energy_conversion_transduction.refrigerant_manifold; energy_conversion_transduction.condenser_coupling_boundary; environmental_interface.outdoor_evaporator; environmental_interface.fan_air_path; plus 25 more	Submodule completeness is judged by deterministic coverage checks, not by a physics simulator.
Interface Graph	passed	Checked class-required module-to-module interfaces before allowing BoM review.	required_links: 4; present_links: 4; missing_links: 0; graph_nodes: 41; graph_edges: 80; shared_interface_edges: 45; missing_required_graph_edges: 0	environmental_interface -> energy_conversion_transduction via refrigerant_loop: present; energy_conversion_transduction -> mass_fluid_transpo	Interfaces prove allocation coverage only; they do not yet validate ratings, thermal margins or dynamic behaviour.

Stage	Status	Summary	Metrics	Evidence	Limitations
Component Candidates	passed	Converted allocated component words into a candidate BoM without supplier prices.	candidate_lines: 124; critical_candidate_lines: 9; admitted_price_lines: 0	rt_process via hydronic_loop: present; control_compute_communication -> energy_conversion_transduction via control_bus: present; sensing_instrumentation -> control_compute_communication via sensor_bus: present	Showing first 30 of 124 component candidates in the stage trace.
				inverter_scroll_compressor: Inverter scroll compressor; compressor_inverter_drive: compressor inverter drive; compressor_mounting_grommet_set: compressor mounting grommet set; crankcase_heater_band: crankcase heater band; electronic_expansion_valve: Electronic expansion valve; eev_stepper_harnesses: EEV stepper harness; plus 24 more	
Architecture Readiness Gate	passed	Architecture validators found no blocker or major issue, so candidate BoM review can start.	ready_for_bom: true; blocking_issue_count: 0; sanity_pass: 3; sanity_warn: 0; sanity_fail: 0	pass COP And Input Power: 3.5 COP / 2.29 kW electric input; pass Refrigerant/Hydraulic Closure: present; pass Safety And Serviceability Interfaces: present	Passing this gate means deterministic architecture checks passed; it is not a certified engineering sign-off.
Sourcing And BoM Admission	warning	No supplier, manufacturer, part number or cost claims have been admitted because no source-backed evidence was provided.	admission_status: not_started; candidate_lines: 124; admitted_lines: 0; unpriced_lines: 124; unpriced_critical_lines: 9; rejected_source_records: 0	none	BoM cost remains intentionally blocked until 9 critical candidate line(s) have explicit source evidence.

Engineering Architecture Review

Ready for BoM review

10 modules, 31 sub-modules, 124 allocated component words.

From	To	Interface	Status	Engineering Reason
environmental_interface	energy_conversion_trans duction	refrigerant_loop	OK	Outdoor evaporator must connect to compressor and expansion circuit over a closed refrigerant loop.
energy_conversion_trans duction	mass_fluid_transport_pr ocess	hydronic_loop	OK	Condenser heat must transfer into the hydronic delivery loop.
control_compute_comm unication	energy_conversion_trans duction	control_bus	OK	Controller must command compressor and expansion valve operation.
sensing_instrumentation	control_compute_comm unication	sensor_bus	OK	Controller must receive pressure and temperature feedback for protection and efficiency.

Interface Graph Summary

Module Nodes	Submodule Nodes	Shared Interface Edges	Required Edges	Missing Required Edges
10	31	45	4	0

Interface Contract Matrix

4/4 required contracts present. 12 shared interface families, 1 local-only declarations.

Interface	From Module	From Carrier	To Module	To Carrier	Status	Reason	Notes
refrigerant_loop	Outdoor Air Interface	Outdoor evaporator	Refrigerant Energy Conversion	Compressor inverter stage; Expansion metering stage; Refrigerant manifold; Condenser coupling boundary	present	Outdoor evaporator must connect to compressor and expansion circuit over a closed refrigerant loop.	Both endpoints expose the required interface through at least one submodule carrier.
hydronic_loop	Refrigerant Energy Conversion	Condenser coupling boundary	Hydronic Heat Delivery	Condenser heat exchanger; Circulation pump group; Hydronic safety group	present	Condenser heat must transfer into the hydronic delivery loop.	Both endpoints expose the required interface through at least one submodule carrier.
control_bus	Control Compute Communication	Main controller; Power driver IO	Refrigerant Energy Conversion	Compressor inverter stage; Expansion metering stage	present	Controller must command compressor and expansion valve operation.	Both endpoints expose the required interface through at least one submodule carrier.
sensor_bus	Sensing Instrumentation	Refrigerant sensing; Water sensing; Ambient sensing	Control Compute Communication	Main controller; Service communications	present	Controller must receive pressure and temperature feedback for protection and efficiency.	Both endpoints expose the required interface through at least one submodule carrier.

Local-only declarations to review later: mechanical_mounts (Monobloc Structure And Containment).

Interface Verification Gate

interface_review_ready

4 required interface row(s): 0 accepted, 4 review-ready, 0 pending evidence, 0 deferred, 0 blocked. Structural pass ratio 1; accepted evidence ratio 0.

Interface	From	To	Carrier Status	Verification Activity	Ledger	Status	Required Action
refrigerant_loop	Outdoor Air Interface	Refrigerant Energy Conversion	both_endpoints_carried	interface_review:environmental_interface:energy_conversion_transduction:refrigerant_loop	pending	review_ready	Collect named interface-review evidence through verification intake.
hydronic_loop	Refrigerant Energy Conversion	Hydronic Heat Delivery	both_endpoints_carried	interface_review:energy_conversion_transduction:mass_fluid_transport_process:hydronic_loop	pending	review_ready	Collect named interface-review evidence through verification intake.
control_bus	Control Compute Communication	Refrigerant Energy Conversion	both_endpoints_carried	interface_review:control_compute_communication:energy_conversion_transduction:control_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
sensor_bus	Sensing Instrumentation	Control Compute Communication	both_endpoints_carried	interface_review:sensing_instrumentation:control_compute_communication:sensor_bus	pending	review_ready	Collect named interface-review evidence through verification intake.

Component Allocation Gate

allocation_ready

31/31 submodules ready; 0 review, 0 blocked. Component words 124; critical parts 9/9; duplicate groups 0. Allocation ratio 1; critical allocation ratio 1.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Refrigerant Energy Conversion	Compressor inverter stage	ready	4	refrigerant_lo op; control_bus; protective_earth	inverter_scroll_compressor	none	Submodule has component candidates and no allocation blocker.
Refrigerant Energy Conversion	Expansion metering stage	ready	4	refrigerant_lo op; control_bus; service_access	electronic_expansion_valve	none	Submodule has component candidates and no allocation blocker.
Refrigerant Energy Conversion	Refrigerant manifold	ready	4	refrigerant_lo op; service_access	none	none	Submodule has component candidates and no allocation blocker.
Refrigerant Energy Conversion	Condenser coupling boundary	ready	4	refrigerant_lo op; hydronic_loop ; sensor_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.
Outdoor Air Interface	Outdoor evaporator	ready	4	refrigerant_lo op; airflow_path; sensor_bus	outdoor_finned_evaporator_coil	none	Submodule has component candidates and no allocation blocker.
Outdoor Air Interface	Fan air path	ready	4	airflow_path; control_bus; protective_earth	variable_speed_fan_assembly	none	Submodule has component candidates and no allocation blocker.
Outdoor Air Interface	Defrost and condensate path	ready	4	condensate_drain; sensor_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.
Hydronic Heat Delivery	Condenser heat exchanger	ready	4	refrigerant_lo op; hydronic_loop ; sensor_bus	brazed_plate_heat_exchanger	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Hydronic Heat Delivery	Circulation pump group	ready	4	hydronic_loop ; control_bus; protective_earth	hydronic_circulation_pump	none	Submodule has component candidates and no allocation blocker.
Hydronic Heat Delivery	Hydronic safety group	ready	4	hydronic_loop ; pressure_relief; service_access	hydronic_safety_valve_kit	none	Submodule has component candidates and no allocation blocker.
Control Compute Communication	Main controller	ready	4	control_bus; sensor_bus; service_port	heat_pump_controller_pcb	none	Submodule has component candidates and no allocation blocker.
Control Compute Communication	Power driver IO	ready	4	control_bus; protective_earth; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Control Compute Communication	Service communications	ready	4	service_port; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Sensing Instrumentation	Refrigerant sensing	ready	4	sensor_bus; refrigerant_loop	refrigerant_sensor_and_pressure_transducers	none	Submodule has component candidates and no allocation blocker.
Sensing Instrumentation	Water sensing	ready	4	sensor_bus; hydronic_loop	none	none	Submodule has component candidates and no allocation blocker.
Sensing Instrumentation	Ambient sensing	ready	4	sensor_bus; airflow_path	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Electrical Power Distribution	Mains input protection	ready	4	mains_supply; protective_earth; service_access	none	none	Submodule has component candidates and no allocation blocker.
Electrical Power Distribution	Internal power rails	ready	4	mains_supply; control_bus; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Electrical Power Distribution	Cable management	ready	4	service_access; sensor_bus; control_bus	none	none	Submodule has component candidates and no allocation blocker.
Monobloc Structure And Containment	Weatherproof cabinet	ready	4	airflow_path; service_access; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Monobloc Structure And Containment	Mounting base	ready	4	mechanical_mounts; condensate_drain	none	none	Submodule has component candidates and no allocation blocker.
Monobloc Structure And Containment	Service access panels	ready	4	service_access	none	none	Submodule has component candidates and no allocation blocker.
Safety Protection	Refrigerant safety	ready	4	refrigerant_loop; sensor_bus; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Safety Protection	Hydronic freeze protection	ready	4	hydronic_loop ; sensor_bus; control_bus	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Safety Protection	Electrical safety	ready	4	protective_earth; mains_supply; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Commissioning	Commissioning ports	ready	4	service_access; refrigerant_loop; hydronic_loop	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Commissioning	Filter service	ready	4	service_access; hydronic_loop ; airflow_path	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Commissioning	Diagnostic access	ready	4	service_port; sensor_bus; control_bus	none	none	Submodule has component candidates and no allocation blocker.
User And Installer Interface	Local display	ready	4	control_bus; service_port	none	none	Submodule has component candidates and no allocation blocker.
User And Installer Interface	Remote thermostat interface	ready	4	control_bus; service_port	none	none	Submodule has component candidates and no allocation blocker.
User And Installer Interface	Alarm and user guidance	ready	4	alarm_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.

Submodule Engineering Gate

submodule_engineering_review_required

0/31 submodules pass; 31 review, 0 blocked. Component words 124; linked requirements 6; carrier contracts 18/18; local-only interfaces 65. Review questions accepted 0/80; verification accepted 0/63; critical unpriced lines 9.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Refrigerant Energy Conversion	Compressor inverter stage	review	4	3	2	2/2	0/4	0/5	1	0.6667	Review 1 local-only interface declaration(s): protective_earth. 4 engineering review question(s) await acceptance. 4 verification activity row(s) await acceptance. 1 evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Refrigerant Energy Conversion	Expansion metering stage	review	4	3	0	2/2	0/4	0/4	1	0.6667	Review 1 local-only interface declaration(s): service_access. 4 engineering review question(s) await acceptance. 3 verification activity row(s)

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Refrigerant Energy Conversion	Refrigerant manifold	review	4	2	0	1/1	0/3	0/2	0	0.6667	Review 1 local-only interface declaration(s): service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Refrigerant Energy Conversion	Condenser coupling boundary	review	4	4	0	2/2	0/4	0/3	0	0.6667	Review 2 local-only interface declaration(s): sensor_bus, service_access. 4 engineering review question(s)

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											await acceptance. 3 verification activity row(s) await accepted evidence.
Outdoor Air Interface	Outdoor evaporator	review	4	3	1	1/1	0/3	0/4	1	0.6667	Review 2 local-only interface declaration(s): airflow_path, sensor_bus. 3 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Outdoor Air Interface	Fan air path	review	4	3	0	0/0	0/2	0/2	1	0.6667	Review 3 local-only interface declaration(s): airflow_path,

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											control_bus, protective_earth. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Outdoor Air Interface	Defrost and condensate path	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): condensate_drain, sensor_bus, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											accepted evidence.
Hydronic Heat Delivery	Condenser heat exchanger	review	4	3	1	1/1	0/3	0/4	1	0.6667	Review 2 local-only interface declaration(s): refrigerant_loop, sensor_bus. 3 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Hydronic Heat Delivery	Circulation pump group	review	4	3	0	1/1	0/3	0/3	1	0.6667	Review 2 local-only interface declaration(s): control_bus, protective_earth. 3 engineering review question(s)

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											n(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Hydronic Heat Delivery	Hydronic safety group	review	4	3	0	1/1	0/3	0/3	1	0.6667	Review 2 local-only interface declaration(s): pressure_relief, service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											cost/manufactur er/MPN evidence.
Control Compute Communication	Main controller	review	4	3	1	2/2	0/4	0/5	1	0.6667	Review 1 local-only interface declaration(s): service_port. 4 engineering review question(s) await acceptance. 4 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.
Control Compute Communication	Power driver IO	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): protective_earth, alarm_bus. 3 engineering review question(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											acceptance. 2 verification activity row(s) await accepted evidence.
Control Compute Communication	Service communications	review	4	2	0	1/1	0/3	0/2	0	0.6667	Review 1 local-only interface declaration(s): service_port. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Sensing Instrumentation	Refrigerant sensing	review	4	2	1	1/1	0/3	0/4	1	0.6667	Review 1 local-only interface declaration(s): refrigerant_loop. 3 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Sensing Instrumentation	Water sensing	review	4	2	0	1/1	0/3	0/2	0	0.6667	Review 1 local-only interface declaration(s): hydronic_loop. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Sensing Instrumentation	Ambient sensing	review	4	2	0	1/1	0/3	0/2	0	0.6667	Review 1 local-only interface declaration(s): airflow_path. 3 engineering review question(s) await acceptance. 2 verification activity

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											row(s) await accepted evidence.
Electrical Power Distribution	Mains input protection	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): mains_supply, protective_earth, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Electrical Power Distribution	Internal power rails	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): mains_supply, control_bus, protective_earth. 2 engineering review question(s) await acceptance. 1 verification

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											activity row(s) await accepted evidence.
Electrical Power Distribution	Cable management	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): service_access, sensor_bus, control_bus. 2 engineer review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Monobloc Structure And Containment	Weather proof cabinet	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): airflow_path, service_access, protective_earth. 2 engineer review question(s) await acceptance. 1 verification activity

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											row(s) await accepted evidence.
Monobloc Structure And Containment	Mounting base	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): mechanical_mounts, condensate_drain. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Monobloc Structure And Containment	Service access panels	review	4	1	0	0/0	0/2	0/1	0	0.6667	Review 1 local-only interface declaration(s): service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											evidence.
Safety Protection	Refrigerant safety	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): refrigerant_loop, sensor_bus, alarm_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Safety Protection	Hydronic freeze protection	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): hydronic_loop, sensor_bus, control_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											evidence.
Safety Protection	Electrical safety	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): protective_earth, mains_supply, alarm_buss. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Maintenance And Commissioning	Commissioning ports	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): service_access, refrigerant_loop, hydronic_loop. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await acceptance

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											d evidence.
Maintenance And Commissioning	Filter service	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): service_access, hydronic_loop, airflow_path. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Maintenance And Commissioning	Diagnostic access	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): service_port, sensor_bus, control_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											evidence.
User And Installer Interface	Local display	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): control_bus, service_port. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
User And Installer Interface	Remote thermostat interface	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): control_bus, service_port. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
User And Installer	Alarm and user	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Interface	guidance										interface declaration(s): alarm_bus, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

Module Engineering Gate

module_engineering_review_required

0/10 modules pass; 10 review, 0 blocked. Submodules/components: 31/124. Interface carriers 8/8; review questions accepted 0/76; critical unpriced lines 9 across 5 module(s).

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Refrigeran t Energy Conversio n	review	4	16	2	3/3	0/14	2	0.86	2 critical BoM line(s) need source- backed cost evidence. 14 engineerin g review question(s) await acceptanc e. 7 assumptio n row(s) need review or source evidence.
Outdoor Air Interface	review	3	12	1	1/1	0/10	2	0.86	2 critical BoM line(s) need source- backed cost evidence. 10 engineerin g review question(s) await acceptanc e. 6 assumptio n row(s) need review or source evidence.
Hydronic Heat Delivery	review	3	12	1	1/1	0/11	3	0.86	3 critical BoM line(s) need source- backed cost evidence. 11 engineerin g review question(s) await acceptanc e. 7 assumptio n row(s) need review or

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									source evidence.
Control Compute Communic ation	review	3	12	1	2/2	0/11	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 11 engineerin g review question(s) await acceptanc e. 5 assumptio n row(s) need review or source evidence.
Sensing Instrument ation	review	3	12	1	1/1	0/10	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 10 engineerin g review question(s) await acceptanc e. 6 assumptio n row(s) need review or source evidence.
Electrical Power Distributio n	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s) await acceptanc e. 1 assumptio n row(s) need review or source evidence.
Monobloc Structure And	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Containme nt									) await acceptanc e.
Safety Protection	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s) await acceptanc e. 1 assumptio n row(s) need review or source evidence.
Maintenan ce And Commissi oning	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s) await acceptanc e. 2 assumptio n row(s) need review or source evidence.
User And Installer Interface	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s) await acceptanc e. 1 assumptio n row(s) need review or source evidence.

Module Review Roll-Up

5 ready, 5 sourcing-blocked, 0 needing attention.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Refrigerant Energy Conversion	sourcing	4	16	3/3	thermal_output_kw; cop	inverter_scroll_compressor; electronic_expansion_valve	2 critical line(s) need source-backed evidence before BoM pass.
Outdoor Air Interface	sourcing	3	12	1/1	thermal_output_kw	outdoor_finned_evaporator_coil; variable_speed_fan_assembly	2 critical line(s) need source-backed evidence before BoM pass.
Hydronic Heat Delivery	sourcing	3	12	1/1	thermal_output_kw	brazed_plate_heat_exchanger; hydronic_circulation_pump; hydronic_safety_valve_kit	3 critical line(s) need source-backed evidence before BoM pass.
Control Compute Communication	sourcing	3	12	2/2	cop	heat_pump_controller_pcb	1 critical line(s) need source-backed evidence before BoM pass.
Sensing Instrumentation	sourcing	3	12	1/1	cop	refrigerant_sensor_and_pressure_transducers	1 critical line(s) need source-backed evidence before BoM pass.
Electrical Power Distribution	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.
Monobloc Structure And Containment	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.
Safety Protection	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.
Maintenance And	ready	3	12	0/0	none	none	No critical blockers; 12

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Commissioning							candidate line(s) remain for later sourcing.
User And Installer Interface	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.

Engineering Review Pack

69 reviewer questions: 0 accepted, 47 ready, 22 needing review, 0 blocked. Coverage: 10 modules, 31 submodules, 4 interfaces, 2 calculations, 22 assumptions.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
assumption_resolution	major	needs review	EN-14511	Can the EN-14511 assumption be accepted for architecture review, or must it be replaced with evidence?	Capacity, COP and operating-point test evidence.	none	none	none	none
assumption_resolution	major	needs review	EN-378	Can the EN-378 assumption be accepted for architecture review, or must it be replaced with evidence?	Refrigerant charge, pressure protection and safety assessment.	none	none	none	none
assumption_resolution	major	needs review	EN-60335-2-40	Can the EN-60335-2-40 assumption be accepted for architecture review, or must it be replaced with evidence?	Electrical safety, leakage, temperature and abnormal-operation evidence.	none	none	none	none
assumption_resolution	major	needs review	Heat pump controller PCB	Can the Heat pump controller PCB assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	control_compute_communication	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
assumptio n_resoluti on	major	needs review	Inverter scroll compressor	Can the Inverter scroll compressor assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	energy_conversion_transduction	none	none	none
assumptio n_resoluti on	major	needs review	Electronic expansion valve	Can the Electronic expansion valve assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	energy_conversion_transduction	none	none	none
assumptio n_resoluti on	major	needs review	Variable-speed fan assembly	Can the Variable-speed fan assembly assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	environmental_interface	none	none	none
assumptio n_resoluti on	major	needs review	Outdoor finned evaporator coil	Can the Outdoor finned evaporator coil assumption be accepted for architecture review, or must it be replaced	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	environmental_interface	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				with evidence?					
assumption_resolution	major	needs review	Hydronic circulation pump	Can the Hydronic circulation pump assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	mass_fluid_transport_process	none	none	none
assumption_resolution	major	needs review	Brazed plate heat exchanger	Can the Brazed plate heat exchanger assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	mass_fluid_transport_process	none	none	none
assumption_resolution	major	needs review	Hydronic safety valve kit	Can the Hydronic safety valve kit assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	mass_fluid_transport_process	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
assumption_resolution	major	needs review	Refrigerant sensor and pressure transducers	Can the Refrigerant sensor and pressure transducers assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	sensing_instrumentation	none	none	none
assumption_resolution	major	needs review	control_compute_communication -> energy_conversion_transduction	Can the control_compute_communication -> energy_conversion_transduction assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	control_buses	none	none
assumption_resolution	major	needs review	energy_conversion_transduction -> mass_fluid_transport_process	Can the energy_conversion_transduction -> mass_fluid_transport_process assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	hydraulic_loop	none	none
assumption_resolution	major	needs review	environmental_interface -> energy_co	Can the environmental_interface ->	Review the interface carrier,	none	refrigerant_loop	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
			nversion_transduction	energy_conversion_transduction assumption be accepted for architecture review, or must it be replaced with evidence?	direction, capacity, safety margin and failure mode before detailed design or sourcing.				
assumption_resolution	major	needs review	sensing_instrumentation -> control_compute_communication	Can the sensing_instrumentation -> control_compute_communication assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	sensor_buses	none	none
assumption_resolution	major	needs review	Estimated CAPEX	Can the Estimated CAPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumption_resolution	major	needs review	Annual useful heat output	Can the Annual useful heat output assumption be accepted for architecture review, or must it be replaced	Review the formula, input values and operating envelope before treating this metric as a performance claim.	none	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				with evidence?					
assumption_resolution	major	needs review	Estimated annual OPEX	Can the Estimated annual OPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumption_resolution	major	needs review	COP And Input Power	Can the COP And Input Power assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	energy_conversion_transduction; environmental_interface; mass_fluid_transport_process; control_communication; sensing_instrumentation	none	none	none
assumption_resolution	major	needs review	Refrigerant/Hydraulic Closure	Can the Refrigerant/Hydraulic Closure assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	energy_conversion_transduction; environmental_interface; mass_fluid_transport_process; control_communication; sensing_instrumentation	none	none	none
assumption_resolution	major	needs review	Safety And Serviceability Interfaces	Can the Safety And Serviceability Interfaces assumption be accepted for	Supply an engineering calculation , reviewer note, test plan or datasheet -backed rationale	none	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				architectu re review, or must it be replaced with evidence?	for this envelope.				
calculation _envelope	major	ready	Heat pump COP envelope	Does the Heat pump COP envelope calculation support the architectu re envelope, or does the concept need revision?	Confirm EN 14511 or local rating point, flow temperatu re, refrigerant and ambient condition.	energy_co nversion_t ransductio n; control_co mpute_co mmunicati on; sensing_in strumenta tion	none	heat_pum p_cop_env elope	none
calculation _envelope	major	ready	Heat pump electrical input	Does the Heat pump electrical input calculation support the architectu re envelope, or does the concept need revision?	Confirm COP rating condition, compress or input power, pump/fan auxiliaries and defrost allowance.	energy_co nversion_t ransductio n; environme ntal_interf ace; mass_fluid _transport _process; control_co mpute_co mmunicati on; sensing_in strumenta tion	none	heat_pum p_electric al_input_k w	none
interface_ contract	major	ready	Control Compute Communication -> Refrigerant Energy Conversion via control_bu s	Is the control_bu s interface between Control Compute Communication and Refrigerant Energy Conversion correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodul es, capacity limits and any deferred checks.	control_co mpute_co mmunicati on; energy_co nversion_t ransductio n	control_bu s	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_ contract	major	ready	Refrigerant Energy Conversion -> Hydronic Heat	Is the hydronic_l oop interface between Refrigeran	Interface-review evidence naming both endpoint	energy_co nversion_t ransductio n; mass_fluid	hydronic_l oop	none	Both endpoints expose the required interface

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
			Delivery via hydronic loop	t Energy Conversion and Hydronic Heat Delivery correctly owned and physically carried?	owners, carrier submodules, capacity limits and any deferred checks.	_transport _process			through at least one submodule carrier.
interface_contract	major	ready	Outdoor Air Interface - > Refrigerant Energy Conversion via refrigerant loop	Is the refrigerant loop interface between Outdoor Air Interface and Refrigerant Energy Conversion correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	environmental_interface; energy_conversion_transduction	refrigerant_loop	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Sensing Instrumentation -> Control Compute Communication via sensor_buses	Is the sensor_buses interface between Sensing Instrumentation and Control Compute Communication correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	sensing_instrumentation; control_compute_communication	sensor_buses	none	Both endpoints expose the required interface through at least one submodule carrier.
module_allocation	major	ready	Control Compute Communication	Does Control Compute Communication have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	control_compute_communication	control_buses; sensor_buses; service_port; alarm_bus; protective_earth	heatpump_cop_input_power; heatpump_refrigerant_hydraulic_closure	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
module_all ocation	major	ready	Refrigeran t Energy Conversio n	Does Refrigeran t Energy Conversio n have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	energy_co nversion_t ransductio n	refrigerant _loop; hydronic_lo op; control_bu s; protective _earth; service_ac cess; sensor_bu s	heatpump _cop_inpu t_power; heatpump _refrigera nt_hydroni c_closure	none
module_all ocation	major	ready	Outdoor Air Interface	Does Outdoor Air Interface have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	environme ntal_interf ace	refrigerant _loop; airflow_pa th; condensat e_drain; sensor_bu s; control_bu s; protective _earth; service_ac cess	heatpump _cop_inpu t_power; heatpump _refrigera nt_hydroni c_closure	none
module_all ocation	major	ready	User And Installer Interface	Does User And Installer Interface have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	hmi_ergon omics	service_p ort; alarm_bus ; control_bu s; service_ac cess	none	none
module_all ocation	major	ready	Maintenan ce And Commissi oning	Does Maintenan ce And Commissi oning have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	maintenan ce_service ability	service_ac cess; service_p ort; refrigerant _loop; hydronic_lo op; airflow_pa th; sensor_bu s; control_bu s	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
module_all ocation	major	ready	Hydronic Heat Delivery	Does Hydronic Heat Delivery have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	mass_fluid _transport _process	hydronic_loop; pressure_relief; sensor_buses; service_access; refrigerant_loop; control_buses; protective_earth	heatpump_cop_input_power; heatpump_refrigerant_hydronic_closure	none
module_all ocation	major	ready	Electrical Power Distribution	Does Electrical Power Distribution have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	power_distribution	mains_supply; protective_earth; control_buses; service_access; sensor_buses	none	none
module_all ocation	major	ready	Safety Protection	Does Safety Protection have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	safety_protection	pressure_relief; alarm_bus; ; protective_earth; sensor_buses; refrigerant_loop; hydronic_loop; control_buses; mains_supply	none	none
module_all ocation	major	ready	Sensing Instrumentation	Does Sensing Instrumentation have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	sensing_instrumentation	sensor_buses; refrigerant_loop; hydronic_loop; airflow_path	heatpump_cop_input_power; heatpump_refrigerant_hydronic_closure	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
module_all ocation	major	ready	Monobloc Structure And Containme nt	Does Monobloc Structure And Containme nt have the right responsibi lity boundary, submodule split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	structure_ containme nt	airflow_pa th; condensat e_drain; service_ac cess; mechanical_mounts; protective_earth	none	none
submodul e_allocation	major	ready	Control Compute Communic ation / Main controller	Do the componen t candidate s and interfaces assigned to Main controller make engineerin g sense for its stated purpose?	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	control_bu s; sensor_bu s; service_p ort	heatpump _cop_inpu t_power; heatpump _refrigera nt_hydroni c_closure	none
submodul e_allocation	major	ready	Control Compute Communic ation / Power driver IO	Do the componen t candidate s and interfaces assigned to Power driver IO make engineerin g sense for its stated purpose?	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	control_bu s; protective_earth; alarm_bus	heatpump _cop_inpu t_power; heatpump _refrigera nt_hydroni c_closure	none
submodul e_allocation	major	ready	Control Compute Communic ation / Service communic ations	Do the componen t candidate s and interfaces assigned to Service communic ations make engineerin g sense for its	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	service_p ort; sensor_bu s	heatpump _cop_inpu t_power; heatpump _refrigera nt_hydroni c_closure	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				stated purpose?					
submodul e_allocation	major	ready	Refrigeran t Energy Conversion / Compressor inverter stage	Do the componen t candidate s and interfaces assigned to Compressor inverter stage make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	energy_co nversion_t ransductio n	refrigerant _loop; control_bu s; protective _earth	heatpump _cop_inpu t_power; heatpump _refrigera nt_hydroni c_closure	none
submodul e_allocation	major	ready	Refrigeran t Energy Conversion / Condenser coupling boundary	Do the componen t candidate s and interfaces assigned to Condenser coupling boundary make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	energy_co nversion_t ransductio n	refrigerant _loop; hydronic_l oop; sensor_bu s; service_ac cess	heatpump _cop_inpu t_power; heatpump _refrigera nt_hydroni c_closure	none
submodul e_allocation	major	ready	Refrigeran t Energy Conversion / Expansion metering stage	Do the componen t candidate s and interfaces assigned to Expansion metering stage make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	energy_co nversion_t ransductio n	refrigerant _loop; control_bu s; service_ac cess	heatpump _cop_inpu t_power; heatpump _refrigera nt_hydroni c_closure	none
submodul e_allocation	major	ready	Refrigeran t Energy Conversion / Refrigeran t manifold	Do the componen t candidate s and interfaces assigned to	Reviewer note confirming the submodul e role, componen t	energy_co nversion_t ransductio n	refrigerant _loop; service_ac cess	heatpump _cop_inpu t_power; heatpump _refrigera nt_hydroni c_closure	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	Outdoor Air Interface / Defrost and condensate path	Refrigerant manifold make engineering sense for its stated purpose?	allocation and any missing carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Defrost and condensate path make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	condensate_drain; sensor_buses; service_access	heatpump_cop_input_power; heatpump_refrigerant_hydronic_closure	none
submodule_allocation	major	ready	Outdoor Air Interface / Fan air path	Do the component candidate s and interfaces assigned to Fan air path make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	airflow_path; control_buses; protective_earth	heatpump_cop_input_power; heatpump_refrigerant_hydronic_closure	none
submodule_allocation	major	ready	Outdoor Air Interface / Outdoor evaporator	Do the component candidate s and interfaces assigned to Outdoor evaporator make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	refrigerant_loop; airflow_path; sensor_buses	heatpump_cop_input_power; heatpump_refrigerant_hydronic_closure	none
submodule_allocation	major	ready	User And Installer Interface / Alarm and user guidance	Do the component candidate s and interfaces assigned	Reviewer note confirming the submodule role, component	hmi_ergonomics	alarm_bus ; service_access	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				to Alarm and user guidance make engineering sense for its stated purpose?	t allocation and any missing carriers or capacity checks.				
submodul e_allocation	major	ready	User And Installer Interface / Local display	Do the componen t candidate s and interfaces assigned to Local display make engineering sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	hmi_ergon omics	control_bu s; service_p ort	none	none
submodul e_allocation	major	ready	User And Installer Interface / Remote thermosta t interface	Do the componen t candidate s and interfaces assigned to Remote thermosta t interface make engineering sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	hmi_ergon omics	control_bu s; service_p ort	none	none
submodul e_allocation	major	ready	Maintenan ce And Commissi oning / Commissi oning ports	Do the componen t candidate s and interfaces assigned to Commissi oning ports make engineering sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	maintenan ce_service ability	service_ac cess; refrigerant _loop; hydronic_lo op	none	none
submodul e_allocation	major	ready	Maintenan ce And Commissi oning /	Do the componen t candidate	Reviewer note confirming the	maintenan ce_service ability	service_p ort; sensor_bu s;	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	Diagnostic access	s and interfaces assigned to Diagnostic access make engineering sense for its stated purpose?	submodule role, component allocation and any missing carriers or capacity checks.		control_bu s		
			Maintenance And Commissioning / Filter service	Do the component candidate s and interfaces assigned to Filter service make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_service_ability	service_access; hydronic_loop; airflow_path	none	none
			Hydronic Heat Delivery / Circulation pump group	Do the component candidate s and interfaces assigned to Circulation pump group make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	hydronic_loop; control_bu s; protective_earth	heatpump_cop_input_power; heatpump_refrigerant_hydronic_closure	none
submodule_allocation	major	ready	Hydronic Heat Delivery / Condenser heat exchanger	Do the component candidate s and interfaces assigned to Condenser heat exchanger make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	refrigerant_loop; hydronic_loop; sensor_bu s	heatpump_cop_input_power; heatpump_refrigerant_hydronic_closure	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Hydronic Heat Delivery / Hydronic safety group	Do the component candidates and interfaces assigned to Hydronic safety group make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	hydronic_loop; pressure_relief; service_access	heatpump_cop_input_power; heatpump_refrigerant_hydronic_closure	none
submodule_allocation	major	ready	Electrical Power Distribution / Cable management	Do the component candidates and interfaces assigned to Cable management make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	service_access; sensor_buses; control_buses	none	none
submodule_allocation	major	ready	Electrical Power Distribution / Internal power rails	Do the component candidates and interfaces assigned to Internal power rails make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	mains_supply; control_buses; protective_earth	none	none
submodule_allocation	major	ready	Electrical Power Distribution / Mains input protection	Do the component candidates and interfaces assigned to Mains input protection make engineering sense for its	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	mains_supply; protective_earth; service_access	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				stated purpose?					
submodule_allocation	major	ready	Safety Protection / Electrical safety	Do the component candidates and interfaces assigned to Electrical safety make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	protective_earth; mains_supply; alarm_bus	none	none
submodule_allocation	major	ready	Safety Protection / Hydronic freeze protection	Do the component candidates and interfaces assigned to Hydronic freeze protection make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	hydronic_loop; sensor_buses; control_buses	none	none
submodule_allocation	major	ready	Safety Protection / Refrigerant safety	Do the component candidates and interfaces assigned to Refrigerant safety make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	refrigerant_loop; sensor_buses; alarm_bus	none	none
submodule_allocation	major	ready	Sensing Instrumentation / Ambient sensing	Do the component candidates and interfaces assigned to Ambient sensing	Reviewer note confirming the submodule role, component allocation and any	sensing_instrumentation	sensor_buses; airflow_path	heatpump_cop_input_power; heatpump_refrigerant_hydraulic_closure	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	Sensing Instrumentation / Refrigerant sensing	make engineering sense for its stated purpose?	missing carriers or capacity checks.				
				Do the component candidates and interfaces assigned to Refrigerant sensing make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	sensing_instrumentation	sensor_buss; refrigerant_loop	heatpump_cop_input_power; heatpump_refrigerant_hydraulic_closure	none
submodule_allocation	major	ready	Sensing Instrumentation / Water sensing	Do the component candidates and interfaces assigned to Water sensing make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	sensing_instrumentation	sensor_buss; hydronic_loop	heatpump_cop_input_power; heatpump_refrigerant_hydraulic_closure	none
submodule_allocation	major	ready	Monobloc Structure And Containment / Mounting base	Do the component candidates and interfaces assigned to Mounting base make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	structure_containment	mechanical_mounts; condensate_drain	none	none
submodule_allocation	major	ready	Monobloc Structure And Containment / Service access panels	Do the component candidates and interfaces assigned to Service access	Reviewer note confirming the submodule role, component allocation	structure_containment	service_access	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				panels make engineering sense for its stated purpose?	and any missing carriers or capacity checks.				
submodule_allocation	major	ready	Monobloc Structure And Containment / Weatherproof cabinet	Do the component candidates and interfaces assigned to Weatherproof cabinet make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	structure_containment	airflow_path; service_access; protective_earth	none	none

Engineering Sanity Checks

Check	Status	Value	Expected	Interpretation
COP And Input Power	pass	3.5 COP / 2.29 kW electric input	COP 2.5 to 5.5 for plausible air-source heat-pump operating points	A 8 kW thermal target at COP 3.5 implies roughly 2.29 kW electrical input before detailed operating-point curves.
Refrigerant/Hydraulic Closure	pass	present	refrigerant loop, hydronic loop, control bus and sensor bus present	Heat extraction, compression, water-side delivery, control and sensing are connected before BoM sourcing.
Safety And Serviceability Interfaces	pass	present	pressure relief, protective earth, condensate drain and service access present	The architecture names the main safety and service interfaces needed for a monobloc heat-pump review.

Engineering Calculation Ledger

2 calculations: 2 within deterministic envelope, 0 needing engineering review, 0 outside envelope, 0 blocked by missing inputs. These are arithmetic gates, not proof of real-world performance.

Calculation	Status	Formula	Result	Envelope	Interpretation	Evidence Required
Heat pump electrical input	within envelope	thermal_output_kw / cop	2.29 kW	Positive arithmetic check; electrical design still needs compressor map and ambient condition.	2.29 kW ideal electrical input before auxiliaries and defrost.	Confirm COP rating condition, compressor input power, pump/fan auxiliaries and defrost allowance.
Heat pump COP envelope	within envelope	cop	3.5 COP	2.5 to 5.5 COP is plausible before rating-condition proof.	3.5 COP target needs rating-condition evidence.	Confirm EN 14511 or local rating point, flow temperature, refrigerant and ambient condition.

Engineering Assumption Ledger

24 assumptions: 2 brief-supported, 4 model-present, 7 needing engineering review, 11 needing source evidence, 0 blocked. Architecture blockers 0; BoM blockers 11.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Thermal output	brief_requirement	brief supported	no	no	Thermal output = 8 kW is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Coefficient of performance	brief_requirement	brief supported	no	no	Coefficient of performance = 3.5 COP is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Annual useful heat output	derived_metric	review required	no	no	Annual useful heat output uses thermal_output_kw * annual_full_load_hours with medium confidence.	Review the formula, input values and operating envelope before treating this metric as a performance claim.
Estimated CAPEX	derived_metric	source required	no	yes	Estimated CAPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Estimated annual OPEX	derived_metric	source required	no	yes	Estimated annual OPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
COP And Input Power	sanity_envelope	review required	no	no	COP And Input Power currently evaluates as pass against COP 2.5 to 5.5 for plausible air-source heat-pump operating points.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Refrigerant/Hydronic Closure	sanity_envelope	review required	no	no	Refrigerant/Hydronic Closure currently evaluates as pass against refrigerant loop, hydronic loop, control bus and sensor bus present.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Safety And Serviceability Interfaces	sanity_envelope	review required	no	no	Safety And Serviceability Interfaces currently evaluates as pass against pressure relief, protective earth, condensate drain and service access present.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
environmental_interface -> energy_conversion_transduction	interface_closure	model present	no	no	refrigerant_loop is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
energy_conversion_transduction -> mass_fluid_transport_process	interface_closure	model present	no	no	hydronic_loop is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
control_compute_communication -> energy_conversion_transduction	interface_closure	model present	no	no	control_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
sensing_instrumentation -> control_compute_communication	interface_closure	model present	no	no	sensor_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
Inverter scroll compressor	critical_component	source required	no	yes	Inverter scroll compressor is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Electronic expansion valve	critical_component	source required	no	yes	Electronic expansion valve is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Outdoor finned evaporator coil	critical_component	source required	no	yes	Outdoor finned evaporator coil is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Variable-speed fan assembly	critical_component	source required	no	yes	Variable-speed fan assembly is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Brazed plate heat exchanger	critical_component	source required	no	yes	Brazed plate heat exchanger is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Hydronic circulation pump	critical_component	source required	no	yes	Hydronic circulation pump is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Hydronic safety valve kit	critical_component	source required	no	yes	Hydronic safety valve kit is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Heat pump controller PCB	critical_component	source required	no	yes	Heat pump controller PCB is a critical architecture-	Admit supplier, manufacturer, MPN where applicable, unit

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Refrigerant sensor and pressure transducers	critical_component	source required	no	yes	derived component candidate with quantity 1 candidate.	cost, lead time and evidence reference through sourcing intake.
					Refrigerant sensor and pressure transducers is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
EN-14511	compliance_evidence	review required	no	no	Heat pump performance test conditions is relevant to this product class in UK/EU.	Capacity, COP and operating-point test evidence.
EN-378	compliance_evidence	review required	no	no	Refrigerating systems and heat pumps safety is relevant to this product class in UK/EU.	Refrigerant charge, pressure protection and safety assessment.
EN-60335-2-40	compliance_evidence	review required	no	no	Electrical safety for heat pumps and air conditioners is relevant to this product class in UK/EU.	Electrical safety, leakage, temperature and abnormal-operation evidence.

Engineering Verification Plan

30 activities: 19 ready for review, 6 open, 5 blocked.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Refrigerant Energy Conversion architecture allocation	ready	design_review	Refrigerant Energy Conversion	thermal_output_kw; cop	refrigerant_loop; hydronic_loop; control_bus; protective_earth; service_access; sensor_bus	inverter_scroll_compressor; compressor_inverter_drive; compressor_mounting_grommet_set; crankcase_heater_band; electronic_expansion_valve; eev_stepper_harness; liquid_line_filter_drier; sight_glass_indicator; plus 8 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Refrigerant Energy Conversion requirement calculations	ready	calculation	Refrigerant Energy Conversion	thermal_output_kw; cop	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Refrigerant Energy Conversion critical components	blocked	source_evidence	Refrigerant Energy Conversion	thermal_output_kw; cop	none	inverter_scroll_compressor; electronic_expansion_valve	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved	inverter_scroll_compressor electronic_expansion_valve

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	
Review Outdoor Air Interface architecture allocation	ready	design_review	Outdoor Air Interface	thermal_output_kw	refrigerant_loop; airflow_path; condensate_drain; sensor_bus; control_bus; protective_earth; service_access	outdoor_finned_evaporator_coil; coil_temperature_sensor; coil_guard_grille; hydrophilic_fin_coating; variable_speed_fan_assembly; ec_fan_motor_controller; fan_shroud; inlet_debris_screen; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Outdoor Air Interface requirement calculations	ready	calculation	Outdoor Air Interface	thermal_output_kw	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Collect source-backed evidence for Outdoor Air Interface critical components	blocked	source_evidence	Outdoor Air Interface	thermal_output_kw	none	outdoor_finned_evaporator_coil; variable_speed_fan_assembly	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	outdoor_finned_evaporator_coil variable_speed_fan_assembly
Review Hydronic Heat Delivery architecture allocation	ready	design_review	Hydronic Heat Delivery	thermal_output_kw	hydronic_loop; pressure_relief; sensor_bus; service_access; refrigerant_loop; control_bus; protective_earth	brazed_plate_heat_exchanger; plate_exchanger_insulation_jacket; water_outlet_temperature_sensor; refrigerant_outlet_temperature_sensor; hydronic_circulation_pump; pump_isolation_valve_pair; flow_proving_switch; strainer_filter; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Hydronic Heat Delivery requirement calculations	ready	calculation	Hydronic Heat Delivery	thermal_output_kw	none	none	Each linked requirement has a calculation, sanity check or clearly	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	
Collect source-backed evidence for Hydronic Heat Delivery critical components	blocked	source_evidence	Hydronic Heat Delivery	thermal_output_kw	none	brazed_plate_heat_exchanger; hydronic_circulation_pump; hydronic_safety_valve_kit	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	brazed_plate_heat_exchanger hydronic_circulation_pump hydronic_safety_valve_kit
Review Control Compute Communication architecture allocation	ready	design_review	Control Compute Communication	cop	control_bus; sensor_bus; service_port; alarm_bus; protective_earth	heat_pump_controller_pcb; controller_enclosure; real_time_clock_module; non_volatile_fault_memory; compressor_contactor; pump_relay_output; fan_pwm_output_board; safety_chain_input_term	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						minal; plus 4 more	No blocker or major architecture issue is attached to the module.	
Verify Control Compute Communication requirement calculations	ready	calculation	Control Compute Communication	cop	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Control Compute Communication critical components	blocked	source_evidence	Control Compute Communication	cop	none	heat_pump_controller_pcb	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	heat_pump_controller_pcb

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Sensing Instrumentation architecture allocation	ready	design_review	Sensing Instrumentation	cop	sensor_bus; refrigerant_loop; hydronic_loop; airflow_path	refrigerant_sensor_and_pressure_transducers; suction_temperature_probe; discharge_temperature_probe; refrigerant_leak_sensor; flow_temperature_sensor; return_temperature_sensor; flow_meter_cartridge; water_pressure_transducer; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Sensing Instrumentation requirement calculations	ready	calculation	Sensing Instrumentation	cop	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Sensing Instrumentation critical components	blocked	source_evidence	Sensing Instrumentation	cop	none	refrigerant_sensor_and_pressure_transducers	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved	refrigerant_sensor_and_pressure_transducers

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	
Review Electrical Power Distribution architecture allocation	ready	design_review	Electrical Power Distribution	none	mains_supply; protective_earth; control_bus; service_access; sensor_bus	lockable_isolator_switch; mcb_breaker_set; rcd_protection_device; 24_v_dc_power_supply; terminal_rail_assembly; fused_auxiliary_terminal; protective_earth_bar; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Review Monobloc Structure And Containment architecture allocation	ready	design_review	Monobloc Structure And Containment	none	airflow_path; condensate_drain; service_access; mechanical_mounts; protective_earth	galvanised_steel_cabin; powder_coated_access_panel; epdm_door_gasket; ip_rated_fastener_set; anti_vibration_foot_set; base_rail_pier; condensate_fall_spacer; lifting_lug_set; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							issue is attached to the module.	
Review Safety Protection architecture allocation	ready	design_review	Safety Protection	none	pressure_relief; alarm_bus; protective_earth; sensor_bus; refrigerant_loop; hydronic_loop; control_bus; ; mains_supply	high_pressure_cut_out_switch; low_pressure_cut_out_switch; refrigerant_leak_alarm_output; refrigerant_warning_label; anti_freeze_thermostat; backup_implementation_helater_relay; drain_down_instruction_label; glycol_compatibility_tag; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Review Maintenance And Commissioning architecture allocation	ready	design_review	Maintenance And Commissioning	none	service_access; service_port; refrigerant_loop; hydronic_loop; airflow_path; ; sensor_bus; control_bus	refrigerant_service_port_cap_set; hydronic_drain_cock; pressure_gauge_pocket; commissioning_checklist_card; cleanable_y_strainer_basket; coil_cleaning_access_cover; service_intervention_label; spare_gasket_kit; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review User And Installer Interface architecture allocation	ready	design_review	User And Installer Interface	none	service_port; alarm_bus; control_bus; service_access	lcd_status_display; menu_button_membrane; weatherproof_display_window; mode_indicator_led_set; thermostat_input_terminal; modbus_room_controller_link; demand_signal_optoisolator; installer_wiring_label; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify refrigerant_loop interface contract	ready	interface_review	Outdoor Air Interface -> Refrigerant Energy Conversion	none	refrigerant_loop	outdoor_finned_evaporator_coil; coil_temperature_sensor; coil_guard_grille; hydrophilic_fin_coating; inverter_scroll_compressor; compressor_inverter_drive; compressor_mounting_grommet_set; crankcase_heater_banded; plus 12 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify hydronic_loop interface contract	ready	interface_review	Refrigerant Energy Conversion -> Hydronic Heat Delivery	none	hydronic_loop	refrigerant_stub_out_pair; hydronic_stub_out_pair; plate_heat_exchanger_mounting_bracket; condenser_insulation_sl	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical,	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						eeve; brazed_plate_heat_exchanger; plate_exchanger_insulation_jacket; water_outlet_temperature_sensor; refrigerant_outlet_temperature_sensor; plus 8 more	thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	
Verify control_bus interface contract	ready	interface_review	Control Compute Communication -> Refrigerant Energy Conversion	none	control_bus	heat_pump_controller_pcb; controller_enclosure; real_time_clock_module; non_volatile_fault_memory; compressor_contactor; pump_relay_output; fan_pwm_output_board; safety_chain_input_terminal; plus 8 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify sensor_bus interface contract	ready	interface_review	Sensing Instrumentation -> Control Compute Communication	none	sensor_bus	refrigerant_sensor_and_pressure_transducers; suction_temperature_probe; discharge_temperature_probe; refrigerant_leak_sensor; flow_temperature_sensor; return_temperature_sensor; flow_meter_cartridge; water_pressure_transducer; plus 12 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Plan evidence for EN-14511	open	compliance_review	System Compliance	none	none	none	Capacity, COP and operating-point test evidence.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for EN-378	open	compliance_review	System Compliance	none	none	none	Refrigerant charge, pressure protection and safety assessment.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for EN-60335-2-40	open	compliance_review	System Compliance	none	none	none	Electrical safety, leakage, temperature and abnormal-operation evidence.	Compliance evidence not yet collected in scratch prototype.
Review risk control for Refrigerant leak or flammability hazard	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.
Review risk control for Frozen outdoor coil reduces capacity	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk	Risk evidence not yet reviewed in scratch prototype.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							item has an owner before publication.	
Review risk control for Hydronic overpressure or low flow	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.

Verification Evidence Ledger

0 accepted, 0 rejected, 0 deferred, 25 pending, 0 blocked without evidence. Review coverage 0; acceptance ratio 0. Source-evidence activities remain in sourcing intake.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review Refrigerant Energy Conversion architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Refrigerant Energy Conversion requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Refrigerant Energy Conversion critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Outdoor Air Interface architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Outdoor Air Interface requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Outdoor Air Interface critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Hydronic Heat Delivery architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Hydronic Heat Delivery requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Hydronic Heat Delivery critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Control Compute Communication architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Control Compute Communication requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Collect source-backed evidence for Control Compute Communication critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Sensing Instrumentation architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Sensing Instrumentation requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Sensing Instrumentation critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Electrical Power Distribution architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review Monobloc Structure And Containment architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review Safety Protection architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review Maintenance And Commissioning architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review User And Installer Interface architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify refrigerant_loop interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify hydronic_loop interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify control_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Verify sensor_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Plan evidence for EN-14511	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for EN-378	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for EN-60335-2-40	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Refrigerant leak or flammability hazard	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Frozen outdoor coil reduces capacity	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Hydronic overpressure or low flow	pending	compliance_review			Collect reviewer evidence through the verification intake template.

Refrigerant Energy Conversion

Moves roughly 8 kW of useful heat with a target COP near 3.5.

- refrigerant_loop
- hydronic_loop
- control_bus
- protective_earth
- service_access

Compressor inverter stage

Raises refrigerant pressure while modulating thermal output.

Components: Inverter scroll compressor, compressor inverter drive, compressor mounting grommet set, crankcase heater band

Expansion metering stage

Meters refrigerant into the outdoor coil across operating conditions.

Components: Electronic expansion valve, EEV stepper harness, liquid-line filter drier, sight glass indicator

Refrigerant manifold

Routes refrigerant between compressor, evaporator, condenser and service points.

Components: copper refrigerant tube set, service valve pair, brazed joint sleeve set, refrigerant charge label

Condenser coupling boundary

Defines the refrigerant-to-water handoff into the hydronic heat-delivery module.

Components: refrigerant stub-out pair, hydronic stub-out pair, plate heat exchanger mounting bracket, condenser insulation sleeve

Outdoor Air Interface

Extracts ambient heat while managing airflow, frost and condensate.

- refrigerant_loop
- airflow_path
- condensate_drain
- sensor_bus

Outdoor evaporator

Transfers ambient heat into the refrigerant loop.

Components: Outdoor finned evaporator coil, coil temperature sensor, coil guard grille, hydrophilic fin coating

Fan air path

Drives controlled airflow through the coil face.

Components: Variable-speed fan assembly, EC fan motor controller, fan shroud, inlet debris screen

Defrost and condensate path

Controls frost meltwater and drainage away from the outdoor unit.

Components: condensate drain tray, trace heater cable, defrost temperature probe, drain hose barb

Hydronic Heat Delivery

Transfers condenser heat into the building water circuit with protected flow.

- hydronic_loop
- pressure_relief
- sensor_bus
- service_access

Condenser heat exchanger

Transfers refrigerant heat to the water loop.

Components: Brazed plate heat exchanger, plate exchanger insulation jacket, water outlet temperature sensor, refrigerant outlet temperature sensor

Circulation pump group

Maintains water flow through the condenser and heating circuit.

Components: Hydronic circulation pump, pump isolation valve pair, flow proving switch, strainer filter

Hydronic safety group

Limits water pressure and allows commissioning service.

Components: Hydronic safety valve kit, expansion vessel, automatic air vent, fill and drain valve set

Control Compute Communication

Coordinates refrigerant, fan, pump, defrost and user-demand states.

- control_bus
- sensor_bus
- service_port
- alarm_bus

Main controller

Runs heating, hot-water, anti-freeze and defrost state machines.

Components: Heat pump controller PCB, controller enclosure, real-time clock module, non-volatile fault memory

Power driver IO

Switches controlled loads and reads discrete safety inputs.

Components: compressor contactor, pump relay output, fan PWM output board, safety chain input terminal

Service communications

Exposes commissioning and diagnostics to installers.

Components: Modbus service port, USB commissioning adapter, fault LED stack, installer QR code label

Sensing Instrumentation

Measures refrigerant, water and ambient state for safety and efficiency.

- sensor_bus
- refrigerant_loop
- hydronic_loop
- airflow_path

Refrigerant sensing

Measures pressure and temperature around the refrigerant circuit.

Components: Refrigerant sensor and pressure transducers, suction temperature probe, discharge temperature probe, refrigerant leak sensor

Water sensing

Measures water-loop temperatures and flow evidence.

Components: flow temperature sensor, return temperature sensor, flow meter cartridge, water pressure transducer

Ambient sensing

Measures external conditions that affect capacity and defrost.

Components: ambient temperature sensor, ambient humidity sensor, air pressure switch, rain ingress sensor

Electrical Power Distribution

Distributes mains power to compressor, fan, pump, controller and auxiliaries.

- mains_supply
- protective_earth
- control_bus
- service_access

Mains input protection

Provides isolation and over-current protection at the unit boundary.

Components: lockable isolator switch, MCB breaker set, RCD protection device, surge protection device

Internal power rails

Creates protected rails for control and auxiliary loads.

Components: 24 V DC power supply, terminal rail assembly, fused auxiliary terminal, protective earth bar

Cable management

Separates mains, sensor and refrigerant-adjacent wiring.

Components: segregated cable duct, IP-rated cable gland set, EMC ferrite clamp set, wiring diagram label

Monobloc Structure And Containment

Contains the refrigeration system, controls airflow and supports outdoor mounting.

- airflow_path
- condensate_drain
- service_access
- mechanical_mounts

Weatherproof cabinet

Protects refrigerant, electrical and hydronic assemblies outdoors.

Components: galvanised steel cabinet, powder-coated access panel, EPDM door gasket, IP-rated fastener set

Mounting base

Carries compressor mass and isolates installation vibration.

Components: anti-vibration foot set, base rail pair, condensate fall spacer, lifting lug set

Service access panels

Creates access for commissioning and repair without dismantling the unit.

Components: removable service panel, hinged controller cover, quarter-turn latch set, service clearance label

Safety Protection

Protects against refrigerant, pressure, electrical and freeze hazards.

- pressure_relief
- alarm_bus
- protective_earth
- sensor_bus

Refrigerant safety

Detects abnormal refrigerant conditions and defines response paths.

Components: high-pressure cut-out switch, low-pressure cut-out switch, refrigerant leak alarm output, refrigerant warning label

Hydronic freeze protection

Prevents water-loop freeze damage in low ambient conditions.

Components: anti-freeze thermostat, backup immersion heater relay, drain-down instruction label, glycol compatibility tag

Electrical safety

Maintains protection and earthing integrity.

Components: earth continuity test point, touch-safe terminal cover, over-temperature cut-out, safety compliance label

Maintenance And Commissioning

Supports installation, refrigerant service, hydronic flushing and periodic inspection.

- service_access
- service_port
- refrigerant_loop
- hydronic_loop

Commissioning ports

Provides controlled access to refrigerant and water commissioning points.

Components: refrigerant service port cap set, hydronic drain cock, pressure gauge pocket, commissioning checklist card

Filter service

Keeps hydronic and air paths maintainable.

Components: cleanable Y-strainer basket, coil cleaning access cover, service interval label, spare gasket kit

Diagnostic access

Lets technicians verify states without dismantling assemblies.

Components: diagnostic test header, fault code label, data log export button, installer handover record

User And Installer Interface

Presents operating mode, alarms and service instructions to users and installers.

- service_port
- alarm_bus
- control_bus

Local display

Shows heat-pump status and permits basic configuration.

Components: LCD status display, menu button membrane, weatherproof display window, mode indicator LED set

Remote thermostat interface

Connects building demand to heat pump control.

Components: thermostat input terminal, Modbus room controller link, demand signal opto-isolator, installer wiring label

Alarm and user guidance

Communicates faults and safe user actions.

Components: alarm relay output, user quick-start label, fault reset button, QR service documentation label

Bill of Materials

Pricing status: architecture-derived candidate BoM only. Costs, suppliers, manufacturers and part numbers have not been independently sourced.

Total CAPEX BoM: £0

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
Inverter scroll compressor	1	candidate	unpriced	unpriced				
compressor inverter drive	1	candidate	unpriced	unpriced				
compressor mounting grommet set	1	candidate	unpriced	unpriced				
crankcase heater band	1	candidate	unpriced	unpriced				
Electronic expansion valve	1	candidate	unpriced	unpriced				
EEV stepper harness	1	candidate	unpriced	unpriced				
liquid-line filter drier	1	candidate	unpriced	unpriced				
sight glass indicator	1	candidate	unpriced	unpriced				
copper refrigerant tube set	1	candidate	unpriced	unpriced				
service valve pair	1	candidate	unpriced	unpriced				
brazed joint sleeve set	1	candidate	unpriced	unpriced				
refrigerant charge label	1	candidate	unpriced	unpriced				
refrigerant stub-out pair	1	candidate	unpriced	unpriced				
hydronic stub-out pair	1	candidate	unpriced	unpriced				
plate heat exchanger mounting bracket	1	candidate	unpriced	unpriced				
condenser insulation sleeve	1	candidate	unpriced	unpriced				
Outdoor finned evaporator coil	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
coil temperature sensor	1	candidate	unpriced	unpriced				
coil guard grille	1	candidate	unpriced	unpriced				
hydrophilic fin coating	1	candidate	unpriced	unpriced				
Variable-speed fan assembly	1	candidate	unpriced	unpriced				
EC fan motor controller	1	candidate	unpriced	unpriced				
fan shroud	1	candidate	unpriced	unpriced				
inlet debris screen	1	candidate	unpriced	unpriced				
condensate drain tray	1	candidate	unpriced	unpriced				
trace heater cable	1	candidate	unpriced	unpriced				
defrost temperature probe	1	candidate	unpriced	unpriced				
drain hose barb	1	candidate	unpriced	unpriced				
Brazed plate heat exchanger	1	candidate	unpriced	unpriced				
plate exchanger insulation jacket	1	candidate	unpriced	unpriced				
water outlet temperature sensor	1	candidate	unpriced	unpriced				
refrigerant outlet temperature sensor	1	candidate	unpriced	unpriced				
Hydronic circulation pump	1	candidate	unpriced	unpriced				
pump isolation valve pair	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
flow proving switch	1	candidate	unpriced	unpriced				
strainer filter	1	candidate	unpriced	unpriced				
Hydronic safety valve kit	1	candidate	unpriced	unpriced				
expansion vessel	1	candidate	unpriced	unpriced				
automatic air vent	1	candidate	unpriced	unpriced				
fill and drain valve set	1	candidate	unpriced	unpriced				
Heat pump controller PCB	1	candidate	unpriced	unpriced				
controller enclosure	1	candidate	unpriced	unpriced				
real-time clock module	1	candidate	unpriced	unpriced				
non-volatile fault memory	1	candidate	unpriced	unpriced				
compressor contactor	1	candidate	unpriced	unpriced				
pump relay output	1	candidate	unpriced	unpriced				
fan PWM output board	1	candidate	unpriced	unpriced				
safety chain input terminal	1	candidate	unpriced	unpriced				
Modbus service port	1	candidate	unpriced	unpriced				
USB commissioning adapter	1	candidate	unpriced	unpriced				
fault LED stack	1	candidate	unpriced	unpriced				
installer QR code label	1	candidate	unpriced	unpriced				
Refrigerant sensor and	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
pressure transducers								
suction temperature probe	1	candidate	unpriced	unpriced				
discharge temperature probe	1	candidate	unpriced	unpriced				
refrigerant leak sensor	1	candidate	unpriced	unpriced				
flow temperature sensor	1	candidate	unpriced	unpriced				
return temperature sensor	1	candidate	unpriced	unpriced				
flow meter cartridge	1	candidate	unpriced	unpriced				
water pressure transducer	1	candidate	unpriced	unpriced				
ambient temperature sensor	1	candidate	unpriced	unpriced				
ambient humidity sensor	1	candidate	unpriced	unpriced				
air pressure switch	1	candidate	unpriced	unpriced				
rain ingress sensor	1	candidate	unpriced	unpriced				
lockable isolator switch	1	candidate	unpriced	unpriced				
MCB breaker set	1	candidate	unpriced	unpriced				
RCD protection device	1	candidate	unpriced	unpriced				
surge protection device	1	candidate	unpriced	unpriced				
24 V DC power supply	1	candidate	unpriced	unpriced				
terminal rail assembly	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
fused auxiliary terminal	1	candidate	unpriced	unpriced				
protective earth bar	1	candidate	unpriced	unpriced				
segregated cable duct	1	candidate	unpriced	unpriced				
IP-rated cable gland set	1	candidate	unpriced	unpriced				
EMC ferrite clamp set	1	candidate	unpriced	unpriced				
wiring diagram label	1	candidate	unpriced	unpriced				
galvanised steel cabinet	1	candidate	unpriced	unpriced				
powder-coated access panel	1	candidate	unpriced	unpriced				
EPDM door gasket	1	candidate	unpriced	unpriced				
IP-rated fastener set	1	candidate	unpriced	unpriced				
anti-vibration foot set	1	candidate	unpriced	unpriced				
base rail pair	1	candidate	unpriced	unpriced				
condensate fall spacer	1	candidate	unpriced	unpriced				
lifting lug set	1	candidate	unpriced	unpriced				
removable service panel	1	candidate	unpriced	unpriced				
hinged controller cover	1	candidate	unpriced	unpriced				
quarter-turn latch set	1	candidate	unpriced	unpriced				
service clearance label	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
high-pressure cut-out switch	1	candidate	unpriced	unpriced				
low-pressure cut-out switch	1	candidate	unpriced	unpriced				
refrigerant leak alarm output	1	candidate	unpriced	unpriced				
refrigerant warning label	1	candidate	unpriced	unpriced				
anti-freeze thermostat	1	candidate	unpriced	unpriced				
backup immersion heater relay	1	candidate	unpriced	unpriced				
drain-down instruction label	1	candidate	unpriced	unpriced				
glycol compatibility tag	1	candidate	unpriced	unpriced				
earth continuity test point	1	candidate	unpriced	unpriced				
touch-safe terminal cover	1	candidate	unpriced	unpriced				
over-temperature cut-out	1	candidate	unpriced	unpriced				
safety compliance label	1	candidate	unpriced	unpriced				
refrigerant service port cap set	1	candidate	unpriced	unpriced				
hydronic drain cock	1	candidate	unpriced	unpriced				
pressure gauge pocket	1	candidate	unpriced	unpriced				
commissioning checklist card	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
cleanable Y-strainer basket	1	candidate	unpriced	unpriced				
coil cleaning access cover	1	candidate	unpriced	unpriced				
service interval label	1	candidate	unpriced	unpriced				
spare gasket kit	1	candidate	unpriced	unpriced				
diagnostic test header	1	candidate	unpriced	unpriced				
fault code label	1	candidate	unpriced	unpriced				
data log export button	1	candidate	unpriced	unpriced				
installer handover record	1	candidate	unpriced	unpriced				
LCD status display	1	candidate	unpriced	unpriced				
menu button membrane	1	candidate	unpriced	unpriced				
weatherpro of display window	1	candidate	unpriced	unpriced				
mode indicator LED set	1	candidate	unpriced	unpriced				
thermostat input terminal	1	candidate	unpriced	unpriced				
Modbus room controller link	1	candidate	unpriced	unpriced				
demand signal opto-isolator	1	candidate	unpriced	unpriced				
installer wiring label	1	candidate	unpriced	unpriced				
alarm relay output	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
user quick-start label	1	candidate	unpriced	unpriced				
fault reset button	1	candidate	unpriced	unpriced				
QR service documentat ion label	1	candidate	unpriced	unpriced				

Cost And Sourcing

Sourcing Admission

Status	Candidate Lines	Admitted Prices	Unpriced Lines	Unpriced Critical Lines
not_started	124	0	124	9

Critical Sourcing Worklist

Component	Qty	Reason
Inverter scroll compressor	1 candidate	Required class-critical line must be sourced before BoM can pass.
Electronic expansion valve	1 candidate	Required class-critical line must be sourced before BoM can pass.
Outdoor finned evaporator coil	1 candidate	Required class-critical line must be sourced before BoM can pass.
Variable-speed fan assembly	1 candidate	Required class-critical line must be sourced before BoM can pass.
Brazed plate heat exchanger	1 candidate	Required class-critical line must be sourced before BoM can pass.
Hydronic circulation pump	1 candidate	Required class-critical line must be sourced before BoM can pass.
Hydronic safety valve kit	1 candidate	Required class-critical line must be sourced before BoM can pass.
Heat pump controller PCB	1 candidate	Required class-critical line must be sourced before BoM can pass.
Refrigerant sensor and pressure transducers	1 candidate	Required class-critical line must be sourced before BoM can pass.

No rejected source records.

BoM Sourcing Ledger

0/124 priced lines admitted. Critical coverage 0/9. Rejected evidence records: 0. Cost/manufacturer/MPN fields stay blank until evidence is admitted.

Component identity check: no duplicate componentWorldId groups detected.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
Inverter scroll compressor	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
compressor or inverter drive	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
compressor or mounting grommet set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
crankcase heater band	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Electronic expansion valve	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
EEV stepper harness	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
liquid-line filter drier	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
sight glass indicator	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
copper refrigerant tube set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
service valve pair	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
brazed joint sleeve set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
refrigerant charge label	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
refrigerant stub-out pair	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
hydronic stub-out pair	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
plate heat exchanger mounting bracket	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
condenser insulation sleeve	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Outdoor finned evaporator coil	criticalunpriced	unique	critical	unpriced					Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
coil temperature sensor	candidateunpriced	unique	candidate	unpriced					Source after critical lines or

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									when detailed procurement planning starts.
coil guard grille	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
hydrophilic fin coating	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Variable-speed fan assembly	criticalunpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
EC fan motor controller	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
fan shroud	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
inlet debris screen	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
condensate drain tray	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
trace heater cable	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
defrost temperature probe	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
drain hose barb	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Brazed plate heat exchanger	critical unpriced	unique	critical	unpriced					Collect admissible supplier/manufacturer/MPN/unit-cost evidence before

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									BoM can pass.
plate exchanger insulation jacket	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
water outlet temperature sensor	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
refrigerant outlet temperature sensor	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Hydronic circulation pump	criticalunpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
pump isolation valve pair	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
flow proving switch	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
strainer filter	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Hydronic safety valve kit	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
expansion vessel	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
automatic air vent	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
fill and drain valve set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Showing first 40 of 124 ledger rows; export CSV for the full line-by-line ledger.

0 source-backed claims, 36 missing-source claims, 0 provenance violations. Critical missing-source claims: 36. Supplier, manufacturer, MPN and cost fields must appear here with source refs before procurement use.

Line	Claim Field	Status	Value	Source Ref	Next Action
Inverter scroll compressor	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Inverter scroll compressor	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Inverter scroll compressor	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Inverter scroll compressor	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Electronic expansion valve	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Electronic expansion valve	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Electronic expansion valve	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Electronic expansion valve	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Outdoor finned evaporator coil	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
Outdoor finned evaporator coil	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Outdoor finned evaporator coil	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Outdoor finned evaporator coil	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Variable-speed fan assembly	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Variable-speed fan assembly	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Variable-speed fan assembly	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Variable-speed fan assembly	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Brazed plate heat exchanger	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Brazed plate heat exchanger	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
Brazed plate heat exchanger	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Brazed plate heat exchanger	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Hydronic circulation pump	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Hydronic circulation pump	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Hydronic circulation pump	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Hydronic circulation pump	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Hydronic safety valve kit	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Hydronic safety valve kit	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Hydronic safety valve kit	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
Hydronic safety valve kit	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Heat pump controller PCB	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Heat pump controller PCB	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Heat pump controller PCB	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Heat pump controller PCB	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Refrigerant sensor and pressure transducers	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Refrigerant sensor and pressure transducers	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Refrigerant sensor and pressure transducers	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Refrigerant sensor and pressure transducers	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

No sourcing evidence references have been admitted. Manufacturer, MPN and cost fields must remain blank until source-backed evidence is admitted through sourcing intake.

BoM Evidence Trace Matrix

0/124 BoM row(s) are production-eligible; 0 protocol-only, 0 source-reference blocked, 9 critical unsourced. Priced-review rows 0; procurement use no; next row bom-trace:energy_conversion_transduction_compressor_inverter_stage_inverter_scroll_compressor.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
Inverter scroll compressor	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
compressor or inverter drive	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
compressor or mounting grommet set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
crankcase heater band	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Electronic expansion valve	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
EEV stepper harness	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									nt planning starts.
liquid-line filter drier	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
sight glass indicator	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
copper refrigerant tube set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
service valve pair	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
brazed joint sleeve set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
refrigerant charge label	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
refrigerant stub-out pair	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
hydronic stub-out pair	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
plate heat exchanger mounting bracket	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
condenser insulation sleeve	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
Outdoor finned evaporator coil	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
coil temperature sensor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
coil guard grille	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
hydrophilic fin coating	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Variable-speed fan assembly	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
EC fan motor controller	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									nt planning starts.
fan shroud	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
inlet debris screen	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
condensate drain tray	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
trace heater cable	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
defrost temperature probe	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
drain hose barb	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Brazed plate heat exchanger	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
plate exchanger insulation jacket	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
water outlet temperature sensor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
refrigerant outlet temperature sensor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Hydronic circulation pump	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									before BoM can pass.
pump isolation valve pair	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
flow proving switch	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
strainer filter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Hydronic safety valve kit	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
expansion vessel	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
automatic air vent	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
fill and drain valve set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Heat pump controller PCB	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
controller enclosure	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
real-time clock module	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
non-volatile fault memory	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
compressor or contactor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
pump relay output	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
fan PWM output board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
safety chain input terminal	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Modbus service port	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
USB commissioning adapter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 124 BoM trace rows; export CSV for the full trace.

BoM Evidence Closure Plan

124 closure row(s): 9 ready, 0 blocked, 115 deferred. Actions: 9 collect source, 0 repair, 0 replace protocol, 115 candidate deferrals. Procurement blockers 9; priced-review blockers 9; next row bom-close:control_compute_communication_main_controller_heat_pump_controller_pcb.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
1	Heat pump controller PCB	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
2	Inverter scroll compressor	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
3	Electronic expansion valve	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
4	Variable-speed fan assembly	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
5	Outdoor finned evaporator coil	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
6	Hydronic circulation pump	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
7	Brazed plate heat exchanger	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
8	Hydronic safety valve kit	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
9	Refrigerant sensor and pressure transducers	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
10	controller enclosure	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
11	non-volatile fault memory	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
12	real-time clock module	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to	no	no	Keep as candidate-only until critical sourcing is complete or procurement

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
						procurement scope			planning starts.
13	compressor contactor	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
14	fan PWM output board	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
15	pump relay output	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
16	safety chain input terminal	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
17	fault LED stack	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
18	installer QR code label	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
19	Modbus service port	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
20	USB commissioning adapter	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
21	compress or inverter drive	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
22	compress or mounting grommet set	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
23	crankcase heater band	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
24	condenser insulation sleeve	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
25	hydronic stub-out pair	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
26	plate heat exchanger mounting bracket	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
27	refrigerant stub-out pair	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
28	EEV stepper harness	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
29	liquid-line filter drier	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
30	sight glass indicator	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
31	brazed joint sleeve set	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
32	copper refrigerant tube set	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
33	refrigerant charge label	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
34	service valve pair	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
35	condensate drain tray	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
36	defrost temperature probe	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
37	drain hose barb	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
38	trace heater cable	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
39	EC fan motor controller	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
40	fan shroud	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
41	inlet debris screen	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
42	coil guard grille	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
43	coil temperature sensor	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
44	hydrophilic fin coating	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
45	alarm relay output	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
46	fault reset button	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
47	QR service documentation label	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
48	user quick-start label	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
49	LCD status display	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
50	menu button membrane	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 124 BoM closure rows; export CSV for the full queue.

Sourcing Batch Plan

2 batch(es): 1 active, 0 waiting, 1 deferred. Active rows 9; deferred rows 115. Critical source rows 9; repair rows 0; protocol replacements 0. Next batch sourcing-batch:2:critical_source_collection; next item critical_source_collection:1:heat_pump_controller_pcb.

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	Heat pump controller PCB	Heat pump controller PCB; Heat pump controller PCB manufacturer part number; Heat pump controller PCB catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Inverter scroll compressor	Inverter scroll compressor; Inverter scroll compressor manufacturer part number; Inverter scroll compressor catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Electronic expansion valve	Electronic expansion valve; Electronic expansion valve manufacturer part number; Electronic expansion valve catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Variable-speed fan assembly	Variable-speed fan assembly; Variable-speed fan assembly manufacturer part number; Variable-speed fan assembly catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	Outdoor finned evaporator coil	Outdoor finned evaporator coil; Outdoor finned evaporator coil manufacturer part number; Outdoor finned evaporator coil catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Hydronic circulation pump	Hydronic circulation pump; Hydronic circulation pump manufacturer part number; Hydronic circulation pump catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Brazed plate heat exchanger	Brazed plate heat exchanger; Brazed plate heat exchanger manufacturer part number; Brazed plate heat exchanger catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Hydronic safety valve kit	Hydronic safety valve kit; Hydronic safety valve kit manufacturer part number; Hydronic safety valve kit catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	Refrigerant sensor and pressure transducers	Refrigerant sensor and pressure transducers ; Refrigerant sensor and pressure transducers manufacturer part number; Refrigerant sensor and pressure transducers catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
6	Defer Candidate-Only Sourcing	deferred	minor	controller enclosure	controller enclosure; controller enclosure manufacturer part number; controller enclosure catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	non-volatile fault memory	non-volatile fault memory; non-volatile fault memory manufacturer part number; non-volatile fault memory catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	real-time clock module	real-time clock module; real-time clock module manufacturer part number; real-time clock module catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-	deferred	minor	compressor contactor	compressor contactor; compressor	no immediate evidence	no	candidate remains out of

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
	Only Sourcing				contactor manufacturer part number; compressor contactor catalogue price GBP	required unless this candidate is promoted to procurement scope		procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	fan PWM output board	fan PWM output board; fan PWM output board manufacturer part number; fan PWM output board catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	pump relay output	pump relay output; pump relay output manufacturer part number; pump relay output catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	safety chain input terminal	safety chain input terminal; safety chain input terminal manufacturer part number; safety chain input terminal catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	fault LED stack	fault LED stack; fault LED stack manufacturer part number; fault LED stack catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-	deferred	minor	installer QR code label	installer QR code label; installer QR	no immediate evidence	no	candidate remains out of

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
	Only Sourcing				code label manufacturer part number; installer QR code label catalogue price GBP	required unless this candidate is promoted to procurement scope		procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	Modbus service port	Modbus service port; Modbus service port manufacturer part number; Modbus service port catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	USB commissioning adapter	USB commissioning adapter; USB commissioning adapter manufacturer part number; USB commissioning adapter catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	compressor inverter drive	compressor inverter drive; compressor inverter drive manufacturer part number; compressor inverter drive catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

Procurement Readiness Gate

procurement_not_started

0/7 procurement areas pass; 2 review, 5 blocked. Critical production-eligible BoM rows 0/9; procurement-blocking closure rows 9; active sourcing rows 9; deferred sourcing rows 115. Source quality no_sourcing_evidence; evidence authenticity no_evidence; costing costing_not_started; BoM admission candidate_bom_authorized. Can use for procurement: no. Next action: Collect production-quality source evidence for critical BoM rows before procurement use..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture review	review	0.33	architecture_freeze_review_required; structurally ready yes, independent review accepted no.	none	Collect accepted engineering/interface/requirement review evidence before procurement use.
bom evidence trace	blocked	0	0/9 critical rows production-eligible; 9 critical unsourced, 0 protocol-only, 0 source-reference blocked.	energy_conversion_transduction_compressor_inverter_stage_inverter_scroll_compressor: Collect admissible source evidence for this critical line before BoM can pass. energy_conversion_transduction_expansion_metering_stage_electronic_expansion_valve: Collect admissible source evidence for this critical line before BoM can pass. environmental_interface_outdoor_evaporator_outdoor_finned_evaporator_coil: Collect admissible source evidence for this critical line before BoM can pass. environmental_interface_fan_air_path_variable_speed_fan_assembly: Collect admissible source evidence for this critical line before BoM can pass. mass_fluid_transport_process_condenser_heat_exchanger_brazed_plate_heat_exchanger: Collect admissible source evidence for this critical line before BoM can pass. plus 4 more	Collect production-quality source evidence for critical BoM rows before procurement use.
sourcing batch closure	blocked	0.9274	9 procurement-blocking closure row(s); 9 active sourcing row(s), 115 deferred row(s).	bom-close:control_compute_communication_main_controller_heat_pump_controller_pcb: Collect admissible source evidence for this critical line before BoM can pass. bom-close:energy_conve	Work the active sourcing batches until procurement-blocking closure rows are closed.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				rsion_transduction_compressor_inverter_stage_inverter_scroll_compressor: Collect admissible source evidence for this critical line before BoM can pass. bom-close:energy_conversion_transduction_expansion_metering_stage_electronic_expansion_valve: Collect admissible source evidence for this critical line before BoM can pass. bom-close:environmental_interface_fan_air_path_variable_speed_fan_assembly: Collect admissible source evidence for this critical line before BoM can pass. bom-close:environmental_interface_outdoor_evaporator_outdoor_finned_evaporator_coil: Collect admissible source evidence for this critical line before BoM can pass. plus 4 more	
source reference quality	blocked	0	no_sourcing_evidence; 0/0 source-reference row(s) pass; 0 protocol, 0 placeholder.	none	Admit production source evidence before procurement readiness can start.
evidence authenticity	blocked	0	no_evidence; 0/0 evidence rows production-ready; 0 protocol.	No evidence records are present.	Add source and reviewer evidence before procurement readiness can start.
bom costing	blocked	0.14	costing_not_started ; critical priced 0/9; BoM total 0 GBP.	energy_conversion_transduction_compressor_inverter_stage_inverter_scroll_compressor: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_conversion_transduction_expansion_metering_stage_electronic_expansion_valve: Collect admissible	Close costing blockers before procurement use.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. environmental_interface_outdoor_evaporator_outdoor_finned_evaporator_coil: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. environmental_interface_fan_air_path_variable_speed_fan_assembly: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. mass_fluid_transport_process_condenser_heat_exchanger_brazed_plate_heat_exchanger: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 52 more	

prototype procurement policy	review	0	candidate_bom_authorized; display mode candidate_only; prototype procurement use allowed no.	none	Prototype policy keeps procurement use disabled until external procurement review and approval are explicitly added.
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Sourcing Evidence Pack

Evidence packets request source data only. They do not admit supplier, manufacturer, MPN or cost claims.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Inverter scroll compressor	1 candidate	Inverter scroll compressor; Inverter scroll compressor manufacturer part number; Inverter scroll compressor catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Electronic expansion valve	1 candidate	Electronic expansion valve; Electronic expansion valve manufacturer part number; Electronic expansion valve catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Outdoor finned evaporator coil	1 candidate	Outdoor finned evaporator coil; Outdoor finned evaporator coil manufacturer part number; Outdoor finned evaporator coil catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Variable-speed fan assembly	1 candidate	Variable-speed fan assembly; Variable-speed fan assembly manufacturer part number; Variable-speed fan assembly catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Brazed plate heat exchanger	1 candidate	Brazed plate heat exchanger; Brazed plate heat exchanger manufacturer part number; Brazed plate heat exchanger catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Hydronic circulation pump	1 candidate	Hydronic circulation pump; Hydronic circulation pump manufacturer part number; Hydronic circulation pump catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Hydronic safety valve kit	1 candidate	Hydronic safety valve kit; Hydronic safety valve kit manufacturer part number; Hydronic safety valve kit catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Heat pump controller PCB	1 candidate	Heat pump controller PCB; Heat pump controller PCB manufacturer part number; Heat pump controller PCB catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Refrigerant sensor and pressure transducers	1 candidate	Refrigerant sensor and pressure transducers; Refrigerant sensor and pressure transducers manufacturer part number; Refrigerant sensor and pressure transducers catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Cost model status: pending sourcing. Architecture is being reviewed before cost claims are admitted.

Cost Bucket	Estimate
CAPEX	£0
Annual OPEX	£0
NRE	£25,000

No admitted supplier spend yet. Supplier rows appear only after source-backed evidence records are admitted.

Regulatory And Risk

Standard	Title	Jurisdiction	Evidence Required
EN-14511	Heat pump performance test conditions	UK/EU	Capacity, COP and operating-point test evidence.
EN-378	Refrigerating systems and heat pumps safety	UK/EU	Refrigerant charge, pressure protection and safety assessment.
EN-60335-2-40	Electrical safety for heat pumps and air conditioners	UK/EU	Electrical safety, leakage, temperature and abnormal-operation evidence.

Hazard	S	O	D	RPN	Mitigation
Refrigerant leak or flammability hazard	9	3	4	108	Charge limit review, leak detection, ventilation path and ignition-source control.
Frozen outdoor coil reduces capacity	6	5	4	120	Defrost control, coil temperature sensing and condensate drainage.
Hydronic overpressure or low flow	7	3	4	84	Flow proving, safety valve, expansion volume and pump feedback.

Section Issues And Score

bom

- blocker / critical_part_unpriced: Inverter scroll compressor is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Electronic expansion valve is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Outdoor finned evaporator coil is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Variable-speed fan assembly is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Brazed plate heat exchanger is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Hydronic circulation pump is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Hydronic safety valve kit is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Heat pump controller PCB is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Refrigerant sensor and pressure transducers is critical but has no credible unit cost.
- major / low_priced_line_ratio: Only 0% of BoM lines have prices.

Section Score Ledger

Rule: start at 9, subtract 3 per blocker, 1.5 per major, 0.5 per minor, floor at 1. Mean score 7.67. This is a deterministic gate score from current validators, not an external reviewer score or proof of real-world engineering quality. Reviewer acceptance is tracked separately in the verification ledger.

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
executive_summary	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
brief_requirements	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
design_modules	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
bom	9	9/1/0	-28.5	1	below target	blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); major/low_priced_line_ratio (-1.5)	Score is base minus deterministic deductions for attached section issues.
feasibility_notes	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
sources_references	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
							this scored section.
Mean	7.67						